Homework Assignments 2

Chapter 5 Decision Making for Two Samples

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報告概述 / Overview

這次作業總共 10 題，題目全部出自 Montgomery 教科書第 5 章 *Decision Making for Two Samples*。10 題依題型大致可以分成五群：

- 已知變異數的兩平均數推論：5-3、5-5、5-9，使用兩樣本 z 檢定 / 信賴區間 / 樣本數公式。
- 未知變異數的兩平均數推論：5-17 用 pooled t（變異數相等）；5-24 用 Welch t （變異數不等）。
- 配對資料：5-41，配對 t 檢定 + 差值常態性檢查。
- 兩變異數比較：5-59，F 檢定。
- 兩比例推論：5-69（檢定 + power + 樣本數）、5-71（傳統 Wald 下界）、5-73（plus-four / Agresti-Caffo 新 CI）。

每題的處理流程都跟教科書範本一致：先把資料 hard-code 進來、印描述統計、做必要的假設檢查、跑檢定算 P-value，必要時補上 power、樣本數、信賴區間或下界，最後給出實務脈絡的解釋。所有數值依本次作業規範四捨五入到第三位小數，樣本數一律 `ceiling()`。共用工具函式（`Pwr.z.test`、`SplSz.z.test`、`SplSz.z.CI`、`Pwr.t.test`、`TwoPropStats`、`Pwr.prop2.test`、`SplSz.prop2.test`、`get_stats`、`save_normality_plots`)跟套件載入都統一放在腳本最前段，完整 R 原始碼在文末的「附錄 A」一次列出。

> 助教校正：5-69(b)(c) 中提到的 $p_A = 0.4$、$p_B = 0.25$ 是 H_1 下的真實 p（為了算 power / 樣本數而設定），跟資料給的點估計 $\hat{p}_A = 35/100$、$\hat{p}_B = 25/100$ 不同，下方的解題會明確區分。

Exercise 5-3: Bottle Filling Machines

> **5-3.** Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.
> (a) Do you think the engineer is correct? Use the P-value approach.
> (b) If $\alpha = 0.05$, what is the power of the test in part (a) for a true difference in means of 0.04?
> (c) Find a 95% CI on the difference in means. Provide a practical interpretation of this interval.
> (d) Assuming equal sample sizes, what sample size should be used to ensure that $\beta = 0.01$ if the true difference in means is 0.04? Assume that $\alpha = 0.05$.

```r
# ==============================================================================
# EXERCISE 5-3: Bottle Filling Machines  (Section 5-2)
# 兩台機台、已知 sigma1 = 0.020, sigma2 = 0.025；雙尾、alpha = 0.05
# ==============================================================================
print_header("EXERCISE 5-3: BOTTLE FILLING MACHINES")

# 題目資料原樣 hard-code，每組 10 筆
Machine1 <- c(16.03, 16.01,
              16.04, 15.96,
              16.05, 15.98,
              16.05, 16.02,
              16.02, 15.99)

Machine2 <- c(16.02, 16.03,
              15.97, 16.04,
              15.96, 16.02,
              16.01, 16.01,
              15.99, 16.00)

params <- list(sigma  = c(0.020, 0.025),
               n      = c(length(Machine1), length(Machine2)),
               Delta0 = 0,
               alpha  = 0.05,
               conf.level = 0.95)

# --- 1. Basic Stats ---
print_subhead("Sample summary")
print(get_stats(Machine1))
print(get_stats(Machine2))

# 老師常用括號賦值，順手把 xbar、se0、z0 印出來
(xbar  <- c(mean(Machine1), mean(Machine2)))
(se0   <- sqrt(sum(params$sigma^2 / params$n)))
(z0    <- (xbar[1] - xbar[2] - params$Delta0) / se0)

# --- 2. (a) P-value 雙尾 ---
print_subhead("(a) Two-sided z-test (known variances)")
PVal <- 2 * (1 - pnorm(abs(z0)))
cat("H0: mu1  = mu2\n")
cat("H1: mu1 != mu2\n")
cat(sprintf("xbar     = (%.3f, %.3f)\n", xbar[1], xbar[2]))
cat(sprintf("xbar.diff = %.3f\n", xbar[1] - xbar[2]))
cat(sprintf("se0      = %.3f\n",  se0))
```

```r
cat(sprintf("z0       = %.3f\n",  z0))
cat(sprintf("P-value  = %.3f\n",  PVal))
cat("Decision: ",
    if (PVal < params$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")

# 套件交叉驗證：BSDA::z.test 走的是同一條路
zchk <- BSDA::z.test(Machine1, Machine2,
                     sigma.x = params$sigma[1],
                     sigma.y = params$sigma[2],
                     alternative = "two.sided",
                     conf.level  = params$conf.level)
cat(sprintf("[BSDA::z.test]  z = %.3f, p = %.3f\n",
            zchk$statistic, zchk$p.value))

# --- 3. (b) Power for true diff = 0.04 ---
print_subhead("(b) Power for true |mu1 - mu2| = 0.04 (alpha = 0.05)")
delta_b <- 0.04
pwr_b   <- Pwr.z.test(n = params$n, delta = delta_b,
                      sigma = params$sigma, alpha = params$alpha,
                      alt = "two.sided")
cat(sprintf("delta    = %.3f\n", delta_b))
cat(sprintf("Power    = %.3f\n", pwr_b))
cat(sprintf("Beta     = %.3f\n", 1 - pwr_b))

# --- 4. (c) 95% CI for mu1 - mu2 ---
print_subhead("(c) 95% Confidence Interval for mu1 - mu2")
(CritVal <- qnorm(1 - params$alpha/2))
CI <- c((xbar[1] - xbar[2]) - CritVal * se0,
        (xbar[1] - xbar[2]) + CritVal * se0)
cat(sprintf("CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[BSDA::z.test CI] (%.3f , %.3f)\n",
            zchk$conf.int[1], zchk$conf.int[2]))

# --- 5. (d) Sample size for beta = 0.01, alpha = 0.05, delta = 0.04 ---
print_subhead("(d) Equal n so that beta = 0.01 when |mu1 - mu2| = 0.04")
n_req <- SplSz.z.test(delta = 0.04, sigma = params$sigma,
                      alpha = params$alpha, pwr = 0.99, side = 2)
# n_req - 1 跟 n_req 兩個點的 power 都印，方便驗證
pw_m1 <- Pwr.z.test(n_req - 1, delta = 0.04, sigma = params$sigma,
                    alpha = params$alpha, alt = "two.sided")
pw_n  <- Pwr.z.test(n_req,     delta = 0.04, sigma = params$sigma,
                    alpha = params$alpha, alt = "two.sided")
cat(sprintf("Required n (each group) = %d\n", n_req))
cat(sprintf("Power at n-1 (%d) = %.3f , beta = %.3f\n",
            n_req - 1, pw_m1, 1 - pw_m1))
cat(sprintf("Power at n   (%d) = %.3f , beta = %.3f\n",
            n_req,     pw_n,  1 - pw_n))

# Power curve：以 delta 為 x 軸
pwr_grid <- tibble(
  delta = seq(0, 0.10, length.out = 200)
) %>% mutate(
  power = Pwr.z.test(params$n, delta, params$sigma, params$alpha, "two.sided")
)
p_pwr_5_3 <- ggplot(pwr_grid, aes(delta, power)) +
```

```
  geom_line(color = "steelblue", linewidth = 1.1) +
  geom_hline(yintercept = 0.99, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = 0.04, linetype = "dotted",  color = "darkgreen") +
  annotate("text", x = 0.042, y = 0.45, label = "delta = 0.04",
           color = "darkgreen", hjust = 0, size = 4) +
  annotate("text", x = 0.08, y = 0.965, label = "target power = 0.99",
           color = "firebrick", hjust = 0, size = 4) +
  scale_y_continuous(breaks = seq(0, 1, 0.25), limits = c(0, 1.02)) +
  labs(title = "Ex 5-3 Two-sample z-test power curve",
       subtitle = "known sigma1 = 0.020, sigma2 = 0.025; n1 = n2 = 10; alpha = 0.05",
       x = expression(paste("True difference  |", mu[1] - mu[2], "|  (oz)")),
       y = expression(paste("Power  = 1 - ", beta))) +
  theme_minimal(base_size = 13) +
  theme(plot.title  = element_text(face = "bold"),
        axis.title  = element_text(face = "bold"))
ggsave("figures/5-3_power_curve.png", p_pwr_5_3,
       width = 7.5, height = 5.0, dpi = 220)
#>
#> ============================================================
#>  EXERCISE 5-3: BOTTLE FILLING MACHINES
#> ============================================================
#>
#> --- Sample summary ---
#>    n   Mean   Var Std_Dev Std_Error Median
#> 1 10 16.015 0.001    0.03      0.01  16.02
#>    n   Mean   Var Std_Dev Std_Error Median
#> 1 10 16.005 0.001   0.025     0.008  16.01
#> [1] 16.015 16.005
#> [1] 0.01012423
#> [1] 0.9877296
#>
#> --- (a) Two-sided z-test (known variances) ---
#> H0: mu1  = mu2
#> H1: mu1 != mu2
#> xbar     = (16.015, 16.005)
#> xbar.diff = 0.010
#> se0      = 0.010
#> z0       = 0.988
#> P-value  = 0.323
#> Decision:  FAIL TO REJECT H0
#> [BSDA::z.test]  z = 0.988, p = 0.323
#>
#> --- (b) Power for true |mu1 - mu2| = 0.04 (alpha = 0.05) ---
#> delta    = 0.040
#> Power    = 0.977
#> Beta     = 0.023
#>
#> --- (c) 95% Confidence Interval for mu1 - mu2 ---
#> [1] 1.959964
#> CI = (-0.010 , 0.030)
#> [BSDA::z.test CI] (-0.010 , 0.030)
#>
#> --- (d) Equal n so that beta = 0.01 when |mu1 - mu2| = 0.04 ---
#> Required n (each group) = 12
#> Power at n-1 (11) = 0.986 , beta = 0.014
```

```
#> Power at n   (12) = 0.991 , beta = 0.009
```

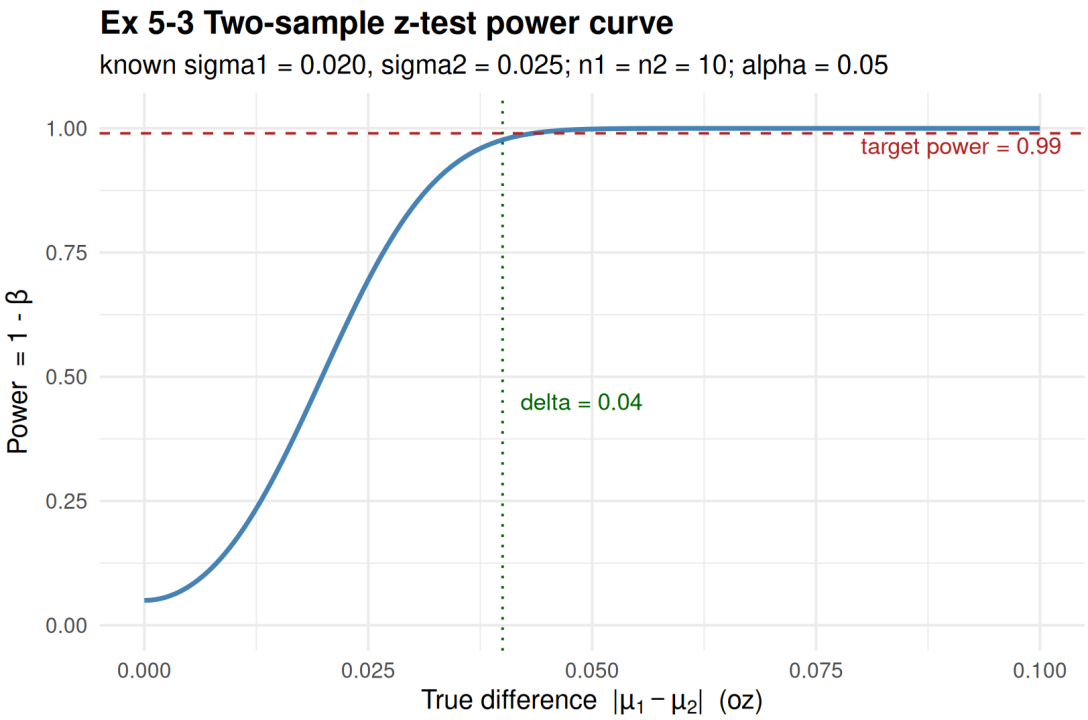

Figure 1: Ex 5-3 Power curve：X 軸是真實平均差 $|\mu_1 - \mu_2|$ (oz)，Y 軸是 power；紅虛線標目標 power $= 0.99$，綠點線標 $\delta = 0.04$。

最終答案 Final Answer

- **(a)** $\bar{x}_1 - \bar{x}_2 = 0.010$、$SE_0 = 0.010$、$z_0 = 0.988$、$P$-value $= 0.323$。$P > 0.05$，**FAIL TO REJECT** H_0：兩台機台的平均填裝量目前沒有顯著差異。
- **(b)** 真實 $|\mu_1 - \mu_2| = 0.04$ 時，power $= 0.977$、$\beta = 0.023$。
- **(c)** 95% CI for $\mu_1 - \mu_2 = (-0.010,\ 0.030)$。0 在 CI 裡，與 (a) 一致。
- **(d)** 要 $\beta = 0.01$、$\alpha = 0.05$、$\delta = 0.04$，$n = \lceil (z_{0.025} + z_{0.01})^2(\sigma_1^2 + \sigma_2^2)/\delta^2 \rceil =$ **12** 每組 ($n = 11$ 時 power $= 0.986$，$n = 12$ 時 power $= 0.991$ 過關)。

Exercise 5-5: Solid-Fuel Propellant Burning Rates

> **5-5.** The burning rates of two different solid-fuel propellants used in aircrew escape systems are being studied. It is known that both propellants have approximately the same standard deviation of burning rate; that is, $\sigma_1 = \sigma_2 = 3$ cm/s. Two random samples of $n_1 = 20$ and $n_2 = 20$ specimens are tested; the sample mean burning rates are $\bar{x}_1 = 18.02$ cm/s and $\bar{x}_2 = 24.37$ cm/s.
>
> (a) Test the hypothesis that both propellants have the same mean burning rate. Use a fixed-level test with $\alpha = 0.05$.
>
> (b) What is the P-value of the test in part (a)?
>
> (c) What is the β-error of the test in part (a) if the true difference in mean burning rate is 2.5 cm/s?
>
> (d) Construct a 95% CI on the difference in means $\mu_1 - \mu_2$. What is the practical meaning of this interval?

```r
# ==============================================================================
# EXERCISE 5-5: Solid-Fuel Propellants  (Section 5-2)
# 已知 sigma1 = sigma2 = 3 cm/s; n1 = n2 = 20; xbar1 = 18.02, xbar2 = 24.37
# ==============================================================================
print_header("EXERCISE 5-5: SOLID-FUEL PROPELLANT BURNING RATES")

# 只有 summary 給；不過 BSDA::zsum.test 剛好吃 summary
xbar  <- c(18.02, 24.37)
sigma <- c(3, 3)
n     <- c(20, 20)
alpha <- 0.05

(se0 <- sqrt(sum(sigma^2 / n)))
(z0  <- (xbar[1] - xbar[2]) / se0)

# --- (a)(b) fixed-level + P-value ---
print_subhead("(a)(b) Two-sided z-test, alpha = 0.05")
(CritVal <- qnorm(1 - alpha/2))
PVal <- 2 * (1 - pnorm(abs(z0)))
cat("H0: mu1  = mu2\n")
cat("H1: mu1 != mu2\n")
cat(sprintf("se0     = %.3f\n", se0))
cat(sprintf("z0      = %.3f\n", z0))
cat(sprintf("CritVal = +/- %.3f\n", CritVal))
cat(sprintf("P-value = %.3f\n", PVal))
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# BSDA::zsum.test 提供同樣的結果，可比對
zsum_chk <- BSDA::zsum.test(mean.x = xbar[1], sigma.x = sigma[1], n.x = n[1],
                            mean.y = xbar[2], sigma.y = sigma[2], n.y = n[2],
                            alternative = "two.sided", conf.level = 1 - alpha)
cat(sprintf("[BSDA::zsum.test]  z = %.3f, p = %.3f\n",
            zsum_chk$statistic, zsum_chk$p.value))

# --- (c) beta when true diff = 2.5 cm/s ---
print_subhead("(c) Beta when true mu1 - mu2 = 2.5 cm/s")
delta_c <- 2.5
pwr_c   <- Pwr.z.test(n = n, delta = delta_c, sigma = sigma,
                      alpha = alpha, alt = "two.sided")
cat(sprintf("delta = %.3f\n",  delta_c))
cat(sprintf("Power = %.3f\n",  pwr_c))
cat(sprintf("Beta  = %.3f\n",  1 - pwr_c))
```

```r
# --- (d) 95% CI for mu1 - mu2 ---
print_subhead("(d) 95% CI for mu1 - mu2")
CI <- c((xbar[1] - xbar[2]) - CritVal * se0,
        (xbar[1] - xbar[2]) + CritVal * se0)
cat(sprintf("CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[zsum.test CI]   (%.3f , %.3f)\n",
            zsum_chk$conf.int[1], zsum_chk$conf.int[2]))

# Beta 區域圖：H0 跟 H1 兩條常態，beta 對應沒被拒絕的部分
# 拿真實 delta = 2.5 的設定畫，能看出 power=0.75 的幾何意義
beta_grid <- seq(-3, 6, length.out = 600)
beta_df <- tibble(
  d        = beta_grid,
  null     = dnorm(beta_grid, mean = 0,       sd = se0),
  alt      = dnorm(beta_grid, mean = 2.5,     sd = se0)
)
crit_lo <- -CritVal * se0
crit_hi <-  CritVal * se0
# 在 H1 下沒被拒絕的區間就是 beta
shade_df <- beta_df %>%
  filter(d >= crit_lo, d <= crit_hi) %>%
  mutate(ymin = 0)
p_beta_5_5 <- ggplot(beta_df, aes(d)) +
  geom_area(aes(y = alt), data = shade_df,
            fill = "orange", alpha = 0.45) +
  geom_line(aes(y = null), color = "steelblue", linewidth = 1) +
  geom_line(aes(y = alt),  color = "firebrick", linewidth = 1) +
  geom_vline(xintercept = c(crit_lo, crit_hi),
             linetype = "dashed", color = "gray30") +
  annotate("text", x = 0,   y = 0.5, label = "H0: mu1 - mu2 = 0",
           color = "steelblue", size = 4, hjust = 0.5) +
  annotate("text", x = 2.5, y = 0.5, label = "H1: mu1 - mu2 = 2.5",
           color = "firebrick", size = 4, hjust = 0.5) +
  annotate("text", x = 1.0, y = 0.12,
           label = sprintf("beta region\n(area = %.3f)", 1 - pwr_c),
           color = "darkorange3", size = 3.8, hjust = 0.5) +
  labs(title = "Ex 5-5 Two-sample z-test: beta region at true delta = 2.5",
       subtitle = "sigma = 3 cm/s, n1 = n2 = 20, alpha = 0.05; two-sided",
       x = expression(paste("Sample mean difference  ",
                            bar(x)[1] - bar(x)[2], "  (cm/s)")),
       y = "Density") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-5_beta_region.png", p_beta_5_5,
       width = 7.8, height = 5.0, dpi = 220)
#>
#> ============================================================
#>  EXERCISE 5-5: SOLID-FUEL PROPELLANT BURNING RATES
#> ============================================================
#> [1] 0.9486833
#> [1] -6.693488
#>
#> --- (a)(b) Two-sided z-test, alpha = 0.05 ---
```

```
#> [1] 1.959964
#> H0: mu1  = mu2
#> H1: mu1 != mu2
#> se0      = 0.949
#> z0       = -6.693
#> CritVal = +/- 1.960
#> P-value = 0.000
#> Decision:  REJECT H0
#> [BSDA::zsum.test]  z = -6.693, p = 0.000
#>
#> --- (c) Beta when true mu1 - mu2 = 2.5 cm/s ---
#> delta = 2.500
#> Power = 0.750
#> Beta  = 0.250
#>
#> --- (d) 95% CI for mu1 - mu2 ---
#> CI = (-8.209 , -4.491)
#> [zsum.test CI]   (-8.209 , -4.491)
```

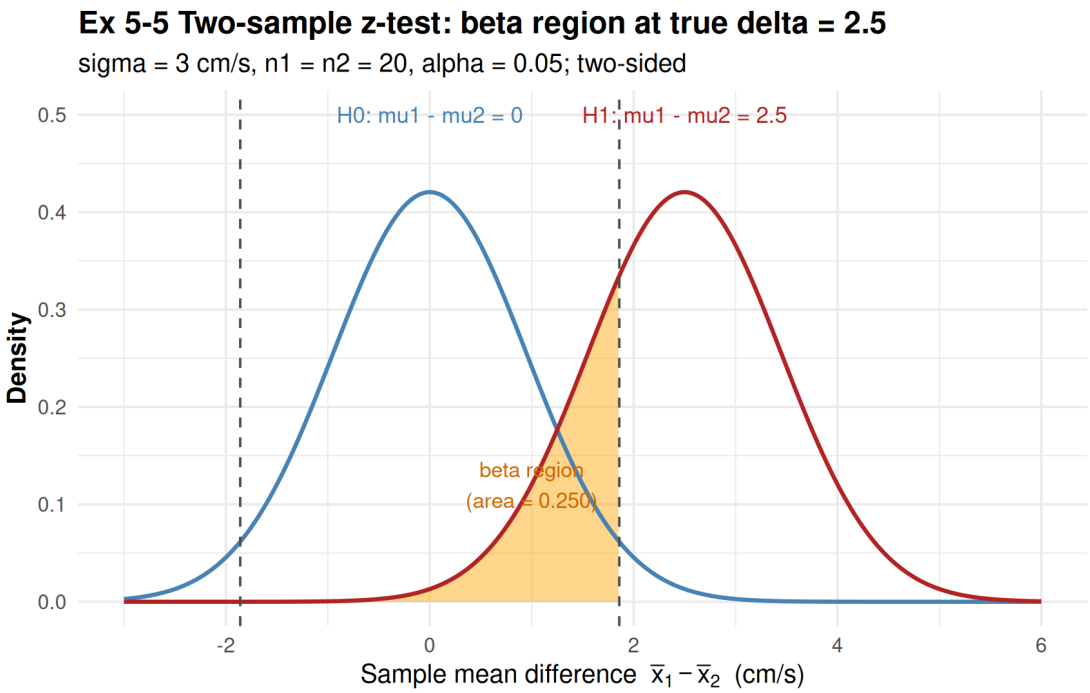

Figure 2: Ex 5-5 在真實 $\delta = 2.5$ 下的 β 區域：藍色為 H_0 抽樣分配、紅色為 H_1 抽樣分配，灰色虛線是雙尾 $\alpha = 0.05$ 的臨界值。橘色區域就是 $\beta = 0.250$。

最終答案 Final Answer

- **(a)(b)** $SE_0 = 0.949$、$z_0 = -6.693$、$|z_0| \gg z_{0.025} = 1.960$，雙尾 P-value ≈ 0（印為 0.000）。$P < 0.05$，**REJECT** H_0：兩種推進劑的平均燃燒速率顯著不同。
- **(c)** 真實 $\delta = 2.5$ 時，power $= 0.750$、$\beta = 0.250$（雙尾、$\alpha = 0.05$）。
- **(d)** 95% CI for $\mu_1 - \mu_2$：$(-8.209,\ -4.491)$。整個區間落在 0 左邊，印證推進劑 2 平均燃燒速率明顯高於推進劑 1，差距約 $4.5 \sim 8.2$ cm/s。

Exercise 5-9: Sample Size for the Difference in Means

> **5-9.** Consider the situation described in Exercise 5-5. What sample size would be required in each population if we wanted the error in estimating the difference in mean burning rates to be less than 4 cm/s with 99% confidence?

```r
# ==============================================================================
# EXERCISE 5-9: Sample size for estimating mu1 - mu2  (Section 5-2)
# 沿用 5-5：sigma1 = sigma2 = 3; equal n; E = 4 cm/s; 99% confidence
# ==============================================================================
print_header("EXERCISE 5-9: SAMPLE SIZE FOR DIFFERENCE IN MEANS")

E <- 4
sigma <- c(3, 3)
conf.level <- 0.99
n_each <- SplSz.z.CI(E = E, sigma = sigma, conf.level = conf.level, side = 2)

# 跟手算公式對一下：n = ((z_{0.005})^2 * (sigma1^2 + sigma2^2)) / E^2
z_half <- qnorm(1 - (1 - conf.level)/2)
manual <- (z_half^2 * sum(sigma^2)) / E^2
cat(sprintf("Confidence level = %.3f\n", conf.level))
cat(sprintf("Margin of error E = %.0f cm/s\n", E))
cat(sprintf("sigma = (%.0f, %.0f)\n", sigma[1], sigma[2]))
cat(sprintf("z_{0.005} = %.3f\n", z_half))
cat(sprintf("Formula value = %.3f -> ceiling = %d (per group)\n",
            manual, n_each))
#>
#> ============================================================
#>  EXERCISE 5-9: SAMPLE SIZE FOR DIFFERENCE IN MEANS
#> ============================================================
#> Confidence level = 0.990
#> Margin of error E = 4 cm/s
#> sigma = (3, 3)
#> z_{0.005} = 2.576
#> Formula value = 7.464 -> ceiling = 8 (per group)
```

最終答案 Final Answer

直接套兩樣本 z 區間的等樣本數公式：

$$n = \left\lceil \frac{z_{0.005}^2\,(\sigma_1^2+\sigma_2^2)}{E^2} \right\rceil = \left\lceil \frac{2.576^2 \cdot (9+9)}{16} \right\rceil = \lceil 7.464 \rceil = \mathbf{8}\ \text{每組}.$$

Exercise 5-17: Single vs Dual Spindle Saw

5-17. An article in *Electronic Components and Technology Conference* (Vol. 52, 2001, pp. 1167–1171) describes a study comparing single versus dual spindle saw processes for copper metallized wafers. A total of 15 devices of each type were measured for the width of the backside chipouts, $\bar{x}_{single} = 66.385$, $s_{single} = 7.895$ and $\bar{x}_{double} = 45.278$, $s_{double} = 8.612$.

(a) Do the sample data support the claim that both processes have the same chip outputs? Assume that both populations are normally distributed and have the same variance. Answer this question by finding and interpreting the P-value for the test.

(b) Construct a 95% two-sided confidence interval on the mean difference in spindle saw process. Compare this interval to the results in part (a).

(c) If the β-error of the test when the true difference in chip outputs is 15 should not exceed 0.1 when $\alpha = 0.05$, what sample sizes must be used?

```r
# ==========================================================================
# EXERCISE 5-17: Single vs Dual Spindle Saw  (Section 5-3)
# Pooled t (假設變異數相同); n1 = n2 = 15
# xbar_s = 66.385, s_s = 7.895;  xbar_d = 45.278, s_d = 8.612
# ==========================================================================
print_header("EXERCISE 5-17: SINGLE vs DUAL SPINDLE SAW")

n     <- c(15, 15)
xbar  <- c(66.385, 45.278)   # single, double
s     <- c(7.895, 8.612)
alpha <- 0.05

# Pooled SD 跟自由度，老師範例都會印
(s_p <- sqrt(((n[1]-1)*s[1]^2 + (n[2]-1)*s[2]^2) / (sum(n) - 2)))
(se  <- s_p * sqrt(sum(1/n)))
(t0  <- (xbar[1] - xbar[2]) / se)
df  <- sum(n) - 2

# --- (a) Two-sided P-value ---
print_subhead("(a) Two-sided pooled t-test")
PVal <- 2 * (1 - pt(abs(t0), df))
cat("H0: mu_single  = mu_double\n")
cat("H1: mu_single != mu_double\n")
cat(sprintf("s_p     = %.3f\n", s_p))
cat(sprintf("se      = %.3f\n", se))
cat(sprintf("t0      = %.3f\n", t0))
cat(sprintf("df      = %d\n",   df))
cat(sprintf("P-value = %.3g\n", PVal))   # 太小就用科學記號
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# 拿 BSDA::tsum.test 對結果
tsum_chk <- BSDA::tsum.test(mean.x = xbar[1], s.x = s[1], n.x = n[1],
                            mean.y = xbar[2], s.y = s[2], n.y = n[2],
                            var.equal = TRUE, alternative = "two.sided",
                            conf.level = 1 - alpha)
cat(sprintf("[BSDA::tsum.test]  t = %.3f, df = %.0f, p = %.3g\n",
            tsum_chk$statistic, tsum_chk$parameter, tsum_chk$p.value))

# --- (b) 95% two-sided CI ---
print_subhead("(b) 95% CI for mu_single - mu_double")
CritVal <- qt(1 - alpha/2, df)
```

```r
CI <- c((xbar[1] - xbar[2]) - CritVal * se,
        (xbar[1] - xbar[2]) + CritVal * se)
cat(sprintf("t_{0.025, %d} = %.3f\n", df, CritVal))
cat(sprintf("CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[tsum.test CI]   (%.3f , %.3f)\n",
            tsum_chk$conf.int[1], tsum_chk$conf.int[2]))

# --- (c) Sample size for beta <= 0.10 when delta = 15 ---
print_subhead("(c) Equal n so that beta <= 0.10 when |mu1 - mu2| = 15")
# 拿 pooled s_p 當 sigma 估計，alpha=0.05，目標 power=0.9，雙尾
# power.t.test() 給 n 是「每組」樣本數，不用再除 2
spl_5_17 <- power.t.test(delta = 15, sd = s_p, sig.level = alpha,
                         power = 0.90, type = "two.sample",
                         alternative = "two.sided")
n_req <- ceiling(spl_5_17$n)
pw_m1 <- Pwr.t.test(n_req - 1, delta = 15, sigma = s_p, alpha = alpha)
pw_n  <- Pwr.t.test(n_req,     delta = 15, sigma = s_p, alpha = alpha)
cat(sprintf("sigma estimate (= s_p) = %.3f\n", s_p))
cat(sprintf("power.t.test raw n     = %.3f -> ceiling = %d (per group)\n",
            spl_5_17$n, n_req))
cat(sprintf("Power at n-1 (%d) = %.3f , beta = %.3f\n",
            n_req - 1, pw_m1, 1 - pw_m1))
cat(sprintf("Power at n   (%d) = %.3f , beta = %.3f\n",
            n_req,     pw_n,  1 - pw_n))

# t 分布拒絕區圖：df=28, 雙尾 alpha=0.05，順手把觀察到的 t0 也標出來
t_grid <- seq(-8, 8, length.out = 700)
t_df_plot <- tibble(t = t_grid, dens = dt(t_grid, df))
crit_t <- qt(1 - alpha/2, df)
reject_lo <- t_df_plot %>% filter(t <= -crit_t)
reject_hi <- t_df_plot %>% filter(t >=  crit_t)
p_t_5_17 <- ggplot(t_df_plot, aes(t, dens)) +
  geom_area(data = reject_lo, fill = "firebrick", alpha = 0.45) +
  geom_area(data = reject_hi, fill = "firebrick", alpha = 0.45) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_vline(xintercept = c(-crit_t, crit_t),
             linetype = "dashed", color = "gray30") +
  geom_vline(xintercept = t0, linetype = "solid",
             color = "darkgreen", linewidth = 1) +
  annotate("text", x = -crit_t, y = 0.35,
           label = sprintf("-t_{0.025, 28}\n= %.3f", -crit_t),
           color = "firebrick", size = 3.8, hjust = 1.05) +
  annotate("text", x = crit_t,  y = 0.35,
           label = sprintf("t_{0.025, 28}\n= %.3f", crit_t),
           color = "firebrick", size = 3.8, hjust = -0.05) +
  annotate("text", x = t0, y = 0.18,
           label = sprintf("observed t0\n= %.3f", t0),
           color = "darkgreen", size = 3.8, hjust = -0.05) +
  labs(title = "Ex 5-17 Pooled t-test: rejection regions and observed t0",
       subtitle = "df = 28, two-sided alpha = 0.05",
       x = "t statistic", y = "Density") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-17_t_rejection.png", p_t_5_17,
```

```
       width = 7.8, height = 5.0, dpi = 220)
#>
#> ============================================================
#>  EXERCISE 5-17: SINGLE vs DUAL SPINDLE SAW
#> ============================================================
#> [1] 8.261282
#> [1] 3.016594
#> [1] 6.996965
#>
#> --- (a) Two-sided pooled t-test ---
#> H0: mu_single  = mu_double
#> H1: mu_single != mu_double
#> s_p     = 8.261
#> se      = 3.017
#> t0      = 6.997
#> df      = 28
#> P-value = 1.31e-07
#> Decision:  REJECT H0
#> [BSDA::tsum.test]  t = 6.997, df = 28, p = 1.31e-07
#>
#> --- (b) 95% CI for mu_single - mu_double ---
#> t_{0.025, 28} = 2.048
#> CI = (14.928 , 27.286)
#> [tsum.test CI]   (14.928 , 27.286)
#>
#> --- (c) Equal n so that beta <= 0.10 when |mu1 - mu2| = 15 ---
#> sigma estimate (= s_p) = 8.261
#> power.t.test raw n     = 7.480 -> ceiling = 8 (per group)
#> Power at n-1 (7) = 0.876 , beta = 0.124
#> Power at n   (8) = 0.921 , beta = 0.079
```

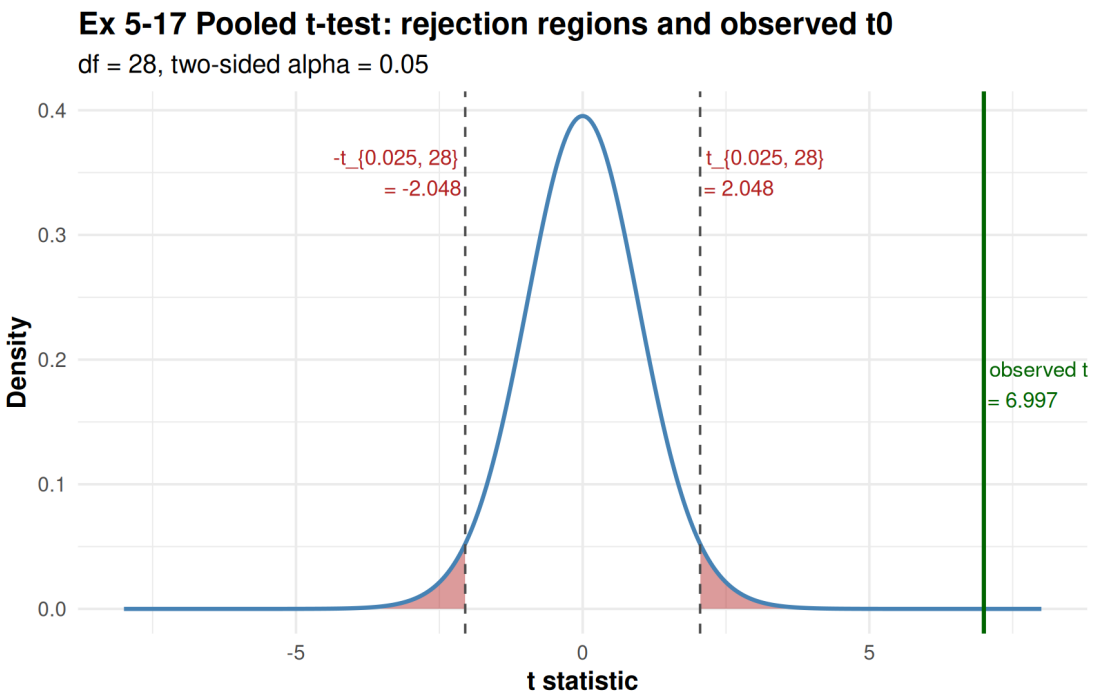

Figure 3: Ex 5-17 pooled t 拒絕區與觀察到的 t_0：X 軸是 t 統計量、Y 軸是密度；灰色虛線是 $\pm t_{0.025,28}$，淺紅色就是雙尾拒絕區，深綠色實線是觀察到的 $t_0 = 6.997$（完全跑到右尾外，所以 P-value 極小）。

最終答案 Final Answer

- **(a)** $s_p = 8.261$、$SE = 3.017$、$t_0 = 6.997$、$df = 28$；雙尾 P-value $\approx 1.31 \times 10^{-7}$，遠小於 0.05 $\Rightarrow$ **REJECT** H_0：single 跟 double 製程的 backside chipout 平均不同。
- **(b)** 95% CI for $\mu_{single} - \mu_{double} = (14.928,\ 27.286)$，整段大於 0，跟 (a) 一致；single 製程的 backside chipout 平均比 double 多出大約 14.9 ~ 27.3 個單位。
- **(c)** 要 $\beta \leq 0.1$、$\delta = 15$、$\alpha = 0.05$，以 s_p 當 σ 估計，$n = \mathbf{8}$ 每組（$n = 7$ 時 power $= 0.876$，$n = 8$ 時 power $= 0.921$）。

Exercise 5-24: Plastic Gear Suppliers

> **5-24.** Two suppliers manufacture a plastic gear used in a laser printer. The impact strength of these gears measured in foot-pounds is an important characteristic. A random sample of 10 gears from supplier 1 results in $\bar{x}_1 = 289.30$ and $s_1 = 22.5$, and another random sample of 16 gears from the second supplier results in $\bar{x}_2 = 321.50$ and $s_2 = 21$.
>
> (a) Is there evidence to support the claim that supplier 2 provides gears with higher mean impact strength? Use the P-value approach, and assume that both populations are normally distributed but the variances are not equal.
>
> (b) Do the data support the claim that the mean impact strength of gears from supplier 2 is at least 25 foot-pounds higher than that of supplier 1? Make the same assumptions as in part (a).
>
> (c) Construct an appropriate 95% CI on the difference in mean impact strength. Use this interval to answer the question posed in part (b).

```r
# ==========================================================================
# EXERCISE 5-24: Plastic Gear Impact Strength  (Section 5-3)
# Welch t (變異數不等); n1 = 10, xbar1 = 289.30, s1 = 22.5
#                       n2 = 16, xbar2 = 321.50, s2 = 21
# ==========================================================================
print_header("EXERCISE 5-24: PLASTIC GEAR SUPPLIERS")

n      <- c(10, 16)
xbar   <- c(289.30, 321.50)   # supplier 1, supplier 2
s      <- c(22.5, 21)
alpha  <- 0.05

# Welch SE 跟 Welch-Satterthwaite df
(se_W <- sqrt(s[1]^2/n[1] + s[2]^2/n[2]))
df_W  <- (s[1]^2/n[1] + s[2]^2/n[2])^2 /
         ((s[1]^2/n[1])^2 / (n[1]-1) + (s[2]^2/n[2])^2 / (n[2]-1))

# --- (a) one-sided test: supplier 2 higher than supplier 1 ---
print_subhead("(a) One-sided Welch t-test, H1: mu2 > mu1")
# 直接寫成 mu2 - mu1 - 0 比較好讀
t0_a <- ((xbar[2] - xbar[1]) - 0) / se_W
PVal_a <- 1 - pt(t0_a, df_W)
cat("H0: mu1 = mu2  (delta = mu2 - mu1 = 0)\n")
cat("H1: mu2 > mu1  (delta > 0)\n")
cat(sprintf("se (Welch)   = %.3f\n", se_W))
cat(sprintf("df (Welch)   = %.3f\n", df_W))
cat(sprintf("t0           = %.3f\n", t0_a))
cat(sprintf("P-value      = %.3f\n", PVal_a))
cat("Decision: ",
    if (PVal_a < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# tsum.test 對一遍
tsum_a <- BSDA::tsum.test(mean.x = xbar[2], s.x = s[2], n.x = n[2],
                          mean.y = xbar[1], s.y = s[1], n.y = n[1],
                          var.equal = FALSE,
                          alternative = "greater", conf.level = 1 - alpha)
cat(sprintf("[BSDA::tsum.test]  t = %.3f, df = %.3f, p = %.3f\n",
            tsum_a$statistic, tsum_a$parameter, tsum_a$p.value))

# --- (b) Test mu2 - mu1 >= 25 ---
print_subhead("(b) Test claim mu2 - mu1 >= 25 (Welch t)")
# 主張要靠資料「站得住腳」，就把它放在 H1
```

```r
Delta0 <- 25
t0_b <- ((xbar[2] - xbar[1]) - Delta0) / se_W
PVal_b <- 1 - pt(t0_b, df_W)
cat("H0: mu2 - mu1  = 25\n")
cat("H1: mu2 - mu1  > 25\n")
cat(sprintf("observed diff = %.3f\n", xbar[2] - xbar[1]))
cat(sprintf("t0            = %.3f\n", t0_b))
cat(sprintf("P-value       = %.3f\n", PVal_b))
cat("Decision: ",
    if (PVal_b < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat("Interpretation: 沒有夠強的證據說『supplier 2 比 supplier 1 平均多 25 foot-pounds 以上』。\n")

# --- (c) 95% CI for mu2 - mu1 ---
print_subhead("(c) 95% CI for mu2 - mu1")
CritVal <- qt(1 - alpha/2, df_W)
CI <- c((xbar[2] - xbar[1]) - CritVal * se_W,
        (xbar[2] - xbar[1]) + CritVal * se_W)
cat(sprintf("t_{0.025, %.2f} = %.3f\n", df_W, CritVal))
cat(sprintf("CI for mu2 - mu1 = (%.3f , %.3f)\n", CI[1], CI[2]))
# 用 CI 回答 (b)：25 在區間裡 -> 不拒絕
cat(sprintf("Is 25 inside the CI? %s -> %s\n",
            ifelse(CI[1] <= 25 && 25 <= CI[2], "Yes", "No"),
            ifelse(CI[1] <= 25 && 25 <= CI[2],
                   "consistent with (b): FAIL TO REJECT",
                   "would REJECT H0 in (b)")))

# CI 圖：把點估計 + 95% CI + 兩條參考線 (0 與 25) 畫在同一條 X 軸上
ci_df_5_24 <- tibble(
  label = "mu2 - mu1",
  est   = xbar[2] - xbar[1],
  lo    = CI[1],
  hi    = CI[2]
)
p_ci_5_24 <- ggplot(ci_df_5_24, aes(y = label)) +
  geom_vline(xintercept = 0,  linetype = "dashed", color = "gray40") +
  geom_vline(xintercept = 25, linetype = "dotdash", color = "firebrick",
             linewidth = 0.8) +
  geom_segment(aes(x = lo, xend = hi, yend = label),
               linewidth = 2, color = "steelblue") +
  geom_point(aes(x = est), size = 4.5, color = "navy") +
  geom_text(aes(x = est, label = sprintf("%.3f", est)),
            vjust = -1.2, size = 4, color = "navy") +
  geom_text(aes(x = lo, label = sprintf("%.3f", lo)),
            vjust = 2.2, size = 3.6, color = "gray30") +
  geom_text(aes(x = hi, label = sprintf("%.3f", hi)),
            vjust = 2.2, size = 3.6, color = "gray30") +
  annotate("text", x = 0,  y = 0.6, label = "0 (test in (a))",
           color = "gray30",   size = 3.6, hjust = -0.05) +
  annotate("text", x = 25, y = 0.6, label = "25 (test in (b))",
           color = "firebrick", size = 3.6, hjust = -0.05) +
  scale_x_continuous(breaks = c(0, 10, 20, 25, 30, 40, 50)) +
  labs(title = "Ex 5-24 Welch 95% CI for mu2 - mu1, with H0 boundaries",
       subtitle = "n1 = 10, n2 = 16; Welch df = 18.226; SE = 8.842",
       x = "Difference in mean impact strength (foot-pounds)", y = NULL) +
  theme_minimal(base_size = 13) +
```

```
  theme(plot.title  = element_text(face = "bold"),
        axis.title  = element_text(face = "bold"),
        axis.text.y = element_text(face = "bold", size = 12))
ggsave("figures/5-24_ci_forest.png", p_ci_5_24,
       width = 8.5, height = 4.2, dpi = 220)
#>
#> ============================================================
#>  EXERCISE 5-24: PLASTIC GEAR SUPPLIERS
#> ============================================================
#> [1] 8.84237
#>
#> --- (a) One-sided Welch t-test, H1: mu2 > mu1 ---
#> H0: mu1 = mu2  (delta = mu2 - mu1 = 0)
#> H1: mu2 > mu1  (delta > 0)
#> se (Welch)    = 8.842
#> df (Welch)    = 18.226
#> t0            = 3.642
#> P-value       = 0.001
#> Decision:  REJECT H0
#> [BSDA::tsum.test]  t = 3.642, df = 18.226, p = 0.001
#>
#> --- (b) Test claim mu2 - mu1 >= 25 (Welch t) ---
#> H0: mu2 - mu1  = 25
#> H1: mu2 - mu1  > 25
#> observed diff = 32.200
#> t0            = 0.814
#> P-value       = 0.213
#> Decision:  FAIL TO REJECT H0
#> Interpretation: 沒有夠強的證據說『supplier 2 比 supplier 1 平均多 25 foot-pounds 以上』。
#>
#> --- (c) 95% CI for mu2 - mu1 ---
#> t_{0.025, 18.23} = 2.099
#> CI for mu2 - mu1 = (13.639 , 50.761)
#> Is 25 inside the CI? Yes -> consistent with (b): FAIL TO REJECT
```

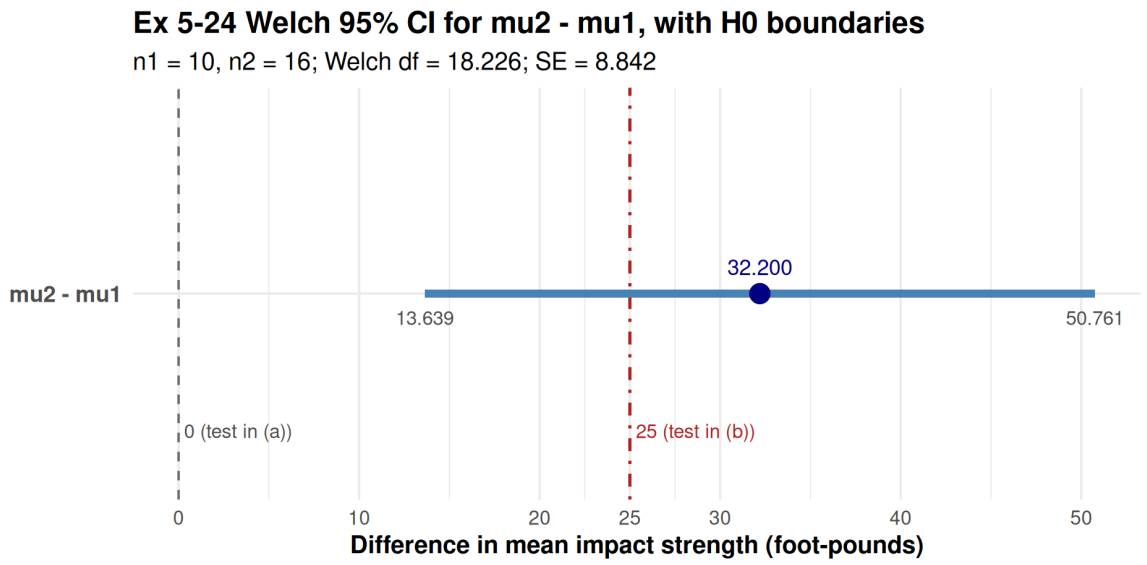

Figure 4: Ex 5-24 Welch 95% CI for $\mu_2 - \mu_1$（藍粗線段），點估計 32.200 在中央。灰色虛線是 (a) 的 $\Delta_0 = 0$，紅色點劃線是 (b) 的 $\Delta_0 = 25$。0 落在 CI 之外（拒絕 (a)）、25 落在 CI 之內（不拒絕 (b)）。

最終答案 Final Answer

- **(a)** Welch $SE = 8.842$、$df = 18.226$、$t_0 = 3.642$、P-value $= 0.001 < 0.05 \Rightarrow$ **REJECT** H_0：supplier 2 的平均衝擊強度明顯較高。
- **(b)** 邊界 $\Delta_0 = 25$，$t_0 = (32.200 - 25)/8.842 = 0.814$、$P$-value $= 0.213 > 0.05 \Rightarrow$ **FAIL TO REJECT** H_0：資料還不夠強到能宣稱「至少高 25 foot-pounds」。
- **(c)** 95% CI for $\mu_2 - \mu_1 = (13.639,\ 50.761)$；25 在 CI 內，跟 (b) 結論一致。實務上 supplier 2 平均比 supplier 1 高 $14 \sim 51$ foot-pounds 都有可能。

Exercise 5-41: Paired Coding Times

> **5-41.** A computer scientist is investigating the usefulness of two different design languages in improving programming tasks. Twelve expert programmers, familiar with both languages, are asked to code a standard function in both languages, and the time in minutes is recorded.
>
> (a) Find a 95% CI on the difference in mean coding times. Is there any indication that one design language is preferable?
>
> (b) Is it reasonable to assume that the difference in coding time is normally distributed? Show evidence to support your answer.

```r
# ==============================================================================
# EXERCISE 5-41: Coding Time, Two Design Languages  (Section 5-4)
# 12 個程式員、配對 t；Coding.diff = Lang1 - Lang2
# ==============================================================================
print_header("EXERCISE 5-41: PAIRED CODING TIMES (TWO DESIGN LANGUAGES)")

Lang1 <- c(17, 16, 21, 14, 18, 24, 16, 14, 21, 23, 13, 18)
Lang2 <- c(18, 14, 19, 11, 23, 21, 10, 13, 19, 24, 15, 20)

# 方向：取 Lang1 - Lang2，表示「語言 1 比語言 2 多用多少分鐘」
Coding.diff <- Lang1 - Lang2
n_p <- length(Coding.diff)
df_p <- n_p - 1
alpha <- 0.05

print_subhead("Summary of paired differences (Lang1 - Lang2)")
print(get_stats(Coding.diff))

# --- (a) 95% CI on mean paired difference ---
print_subhead("(a) 95% CI on mean(Lang1 - Lang2)")
t_paired <- t.test(Lang1, Lang2, paired = TRUE, conf.level = 1 - alpha)
xbar_d <- mean(Coding.diff)
sd_d   <- sd(Coding.diff)
se_d   <- sd_d / sqrt(n_p)
CritVal <- qt(1 - alpha/2, df_p)
CI <- c(xbar_d - CritVal * se_d, xbar_d + CritVal * se_d)
cat(sprintf("xbar_d (Lang1 - Lang2) = %.3f\n", xbar_d))
cat(sprintf("s_d                    = %.3f\n", sd_d))
cat(sprintf("se_d                   = %.3f\n", se_d))
cat(sprintf("t_{0.025, %d}           = %.3f\n", df_p, CritVal))
cat(sprintf("95%% CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[t.test CI]  (%.3f , %.3f)\n",
            t_paired$conf.int[1], t_paired$conf.int[2]))
# 0 在 CI 裡嗎？順手做一下整體解釋
if (CI[1] <= 0 && 0 <= CI[2]) {
  cat("Interpretation: 0 落在 95% CI 內，沒有證據說兩種語言的平均 coding time 不同。\n")
} else {
  cat("Interpretation: 0 落在 95% CI 之外，有證據說兩種語言平均 coding time 不同。\n")
}

# --- (b) Normality of the differences ---
print_subhead("(b) Normality check on the paired differences")
(sw_p <- shapiro.test(Coding.diff))
save_normality_plots(Coding.diff,
                     "Ex 5-41 Coding.diff",
```

```
                      "5-41_diff",
                      x_label = "Coding.diff = Lang1 - Lang2 (minutes)")
# qqPlot 來自 car，老師範例常用
png("figures/5-41_diff_qqplot.png", width = 1100, height = 800, res = 150)
car::qqPlot(Coding.diff,
            main = "Ex 5-41: paired diff Q-Q plot",
            ylab = "Coding.diff (Lang1 - Lang2)")
invisible(dev.off())
cat("Figures saved: figures/5-41_diff_qq.png, figures/5-41_diff_hist.png, ",
    "figures/5-41_diff_qqplot.png\n", sep = "")
#>
#> ============================================================
#>  EXERCISE 5-41: PAIRED CODING TIMES (TWO DESIGN LANGUAGES)
#> ============================================================
#>
#> --- Summary of paired differences (Lang1 - Lang2) ---
#>    n  Mean   Var Std_Dev Std_Error Median
#> 1 12 0.667 8.788   2.964     0.856    1.5
#>
#> --- (a) 95% CI on mean(Lang1 - Lang2) ---
#> xbar_d (Lang1 - Lang2) = 0.667
#> s_d                    = 2.964
#> se_d                   = 0.856
#> t_{0.025, 11}           = 2.201
#> 95% CI = (-1.217 , 2.550)
#> [t.test CI]  (-1.217 , 2.550)
#> Interpretation: 0 落在 95% CI 內，沒有證據說兩種語言的平均 coding time 不同。
#>
#> --- (b) Normality check on the paired differences ---
#>
#>  Shapiro-Wilk normality test
#>
#> data:  Coding.diff
#> W = 0.96165, p-value = 0.8071
#>
#> [1] 5 7
#> Figures saved: figures/5-41_diff_qq.png, figures/5-41_diff_hist.png, figures/
↪  5-41_diff_qqplot.png
```

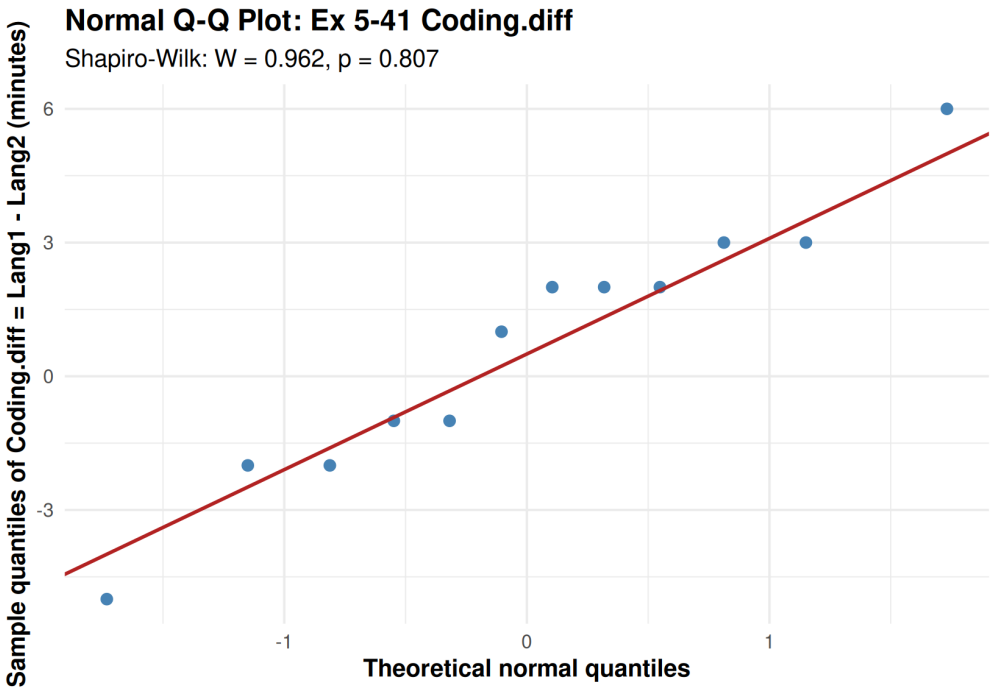

Figure 5: Ex 5-41 配對差值的 Normal Q-Q plot。X 軸是理論常態分位數、Y 軸是樣本分位數（單位：分鐘）；12 個點都貼著紅色參考線，配合 $W = 0.962$、$p = 0.807$ 可以接受差值的常態假設。

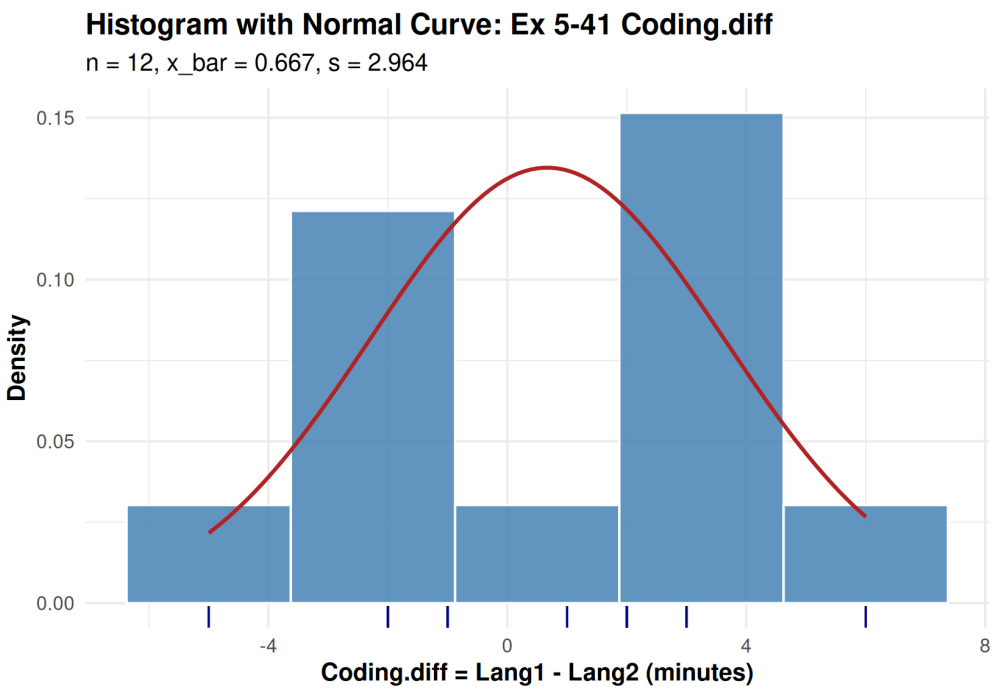

Figure 6: Ex 5-41 配對差值的直方圖與套疊的常態曲線。X 軸是 Coding.diff（分鐘）、Y 軸是 density；底部 rug 把 12 個觀測值的位置點出來。

最終答案 Final Answer

- 方向約定：$D_i = \text{Lang1}_i - \text{Lang2}_i$，$\bar{D} > 0$ 表示語言 1 比語言 2 多用時間。
- **(a)** $\bar{D} = 0.667$、$s_D = 2.964$、$SE_D = 0.856$、$t_{0.025,11} = 2.201$；95% CI $= (-1.217,\ 2.550)$。0 落在 CI 內 $\Rightarrow$ 沒有夠強的證據說某種語言比較快。
- **(b)** Shapiro-Wilk $W = 0.962$、$p = 0.807$；Q-Q plot 上 12 個點都在參考帶內，差值的常態假設可以接受，配

> 對 t 推論是穩當的。

Exercise 5-59: F-Test for Equal Variances (Gear Data)

> **5-59.** Consider the gear impact strength data in Exercise 5-24. Is there sufficient evidence to conclude that the variance of impact strength is different for the two suppliers? Use $\alpha = 0.05$.

```r
# ==============================================================================
# EXERCISE 5-59: Equality of Two Variances  (Section 5-5)
# 沿用 5-24：s1 = 22.5, n1 = 10; s2 = 21, n2 = 16; 雙尾 alpha = 0.05
# ==============================================================================
print_header("EXERCISE 5-59: F-TEST FOR EQUAL VARIANCES (GEAR DATA)")

n     <- c(10, 16)
s     <- c(22.5, 21)
alpha <- 0.05

(f0 <- s[1]^2 / s[2]^2)
df  <- c(n[1] - 1, n[2] - 1)

# 雙尾 P-value：兩端取小的乘 2
PVal <- 2 * min(pf(f0, df[1], df[2]), 1 - pf(f0, df[1], df[2]))
# 兩個臨界值，幫 grader 看
f_low  <- qf(    alpha/2, df[1], df[2])
f_high <- qf(1 - alpha/2, df[1], df[2])

print_subhead("Two-sided F-test, alpha = 0.05")
cat("H0: sigma1^2  = sigma2^2\n")
cat("H1: sigma1^2 != sigma2^2\n")
cat(sprintf("f0      = s1^2 / s2^2 = %.3f\n", f0))
cat(sprintf("df      = (%d, %d)\n", df[1], df[2]))
cat(sprintf("F_{0.025, %d, %d} = %.3f , F_{0.975, %d, %d} = %.3f\n",
            df[1], df[2], f_low, df[1], df[2], f_high))
cat(sprintf("P-value = %.3f\n", PVal))
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")

# 也順手丟 95% CI for sigma1^2 / sigma2^2
LB <- (s[1]^2 / s[2]^2) / qf(1 - alpha/2, df[1], df[2])
UB <- (s[1]^2 / s[2]^2) / qf(    alpha/2, df[1], df[2])
cat(sprintf("95%% CI for sigma1^2 / sigma2^2 = (%.3f , %.3f)\n", LB, UB))
cat("1 落在 CI 裡 -> 跟 P-value 結論一致（不拒絕兩變異數相等）。\n")

# F(9, 15) 分布 + 雙尾拒絕區 + f0 標記
f_grid <- seq(0.001, 6, length.out = 700)
f_df_plot <- tibble(f = f_grid, dens = df(f_grid, df[1], df[2]))
rej_lo_f <- f_df_plot %>% filter(f <= f_low)
rej_hi_f <- f_df_plot %>% filter(f >= f_high)
p_f_5_59 <- ggplot(f_df_plot, aes(f, dens)) +
  geom_area(data = rej_lo_f, fill = "firebrick", alpha = 0.45) +
  geom_area(data = rej_hi_f, fill = "firebrick", alpha = 0.45) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_vline(xintercept = c(f_low, f_high),
             linetype = "dashed", color = "gray30") +
  geom_vline(xintercept = f0, linetype = "solid",
             color = "darkgreen", linewidth = 1) +
  # 三個標籤往畫面右上半排，避開曲線、彼此也不重疊
```

```r
  annotate("text", x = 2.4, y = 0.78,
           label = sprintf("F_{0.025, 9, 15} = %.3f  (left dashed line)", f_low),
           color = "firebrick", size = 4, hjust = 0) +
  annotate("text", x = 2.4, y = 0.68,
           label = sprintf("F_{0.975, 9, 15} = %.3f  (right dashed line)", f_high),
           color = "firebrick", size = 4, hjust = 0) +
  annotate("text", x = 2.4, y = 0.58,
           label = sprintf("observed f0 = %.3f  (solid green line)", f0),
           color = "darkgreen", size = 4, hjust = 0) +
  # tick 拆成整數，避免 1.00 跟 1.15 擠在一起
  scale_x_continuous(breaks = c(0, 1, 2, 3, 4, 5, 6),
                     limits = c(0, 6)) +
  labs(title = "Ex 5-59 Two-sided F-test: rejection regions and observed f0",
       subtitle = "df = (9, 15), alpha = 0.05; f0 = s1^2 / s2^2 = 1.148",
       x = "F statistic", y = "Density") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-59_f_rejection.png", p_f_5_59,
       width = 8.5, height = 5.0, dpi = 220)
#>
#> ============================================================
#>  EXERCISE 5-59: F-TEST FOR EQUAL VARIANCES (GEAR DATA)
#> ============================================================
#> [1] 1.147959
#>
#> --- Two-sided F-test, alpha = 0.05 ---
#> H0: sigma1^2  = sigma2^2
#> H1: sigma1^2 != sigma2^2
#> f0       = s1^2 / s2^2 = 1.148
#> df       = (9, 15)
#> F_{0.025, 9, 15} = 0.265 , F_{0.975, 9, 15} = 3.123
#> P-value = 0.781
#> Decision:  FAIL TO REJECT H0
#> 95% CI for sigma1^2 / sigma2^2 = (0.368 , 4.327)
#> 1 落在 CI 裡 -> 跟 P-value 結論一致（不拒絕兩變異數相等）。
```

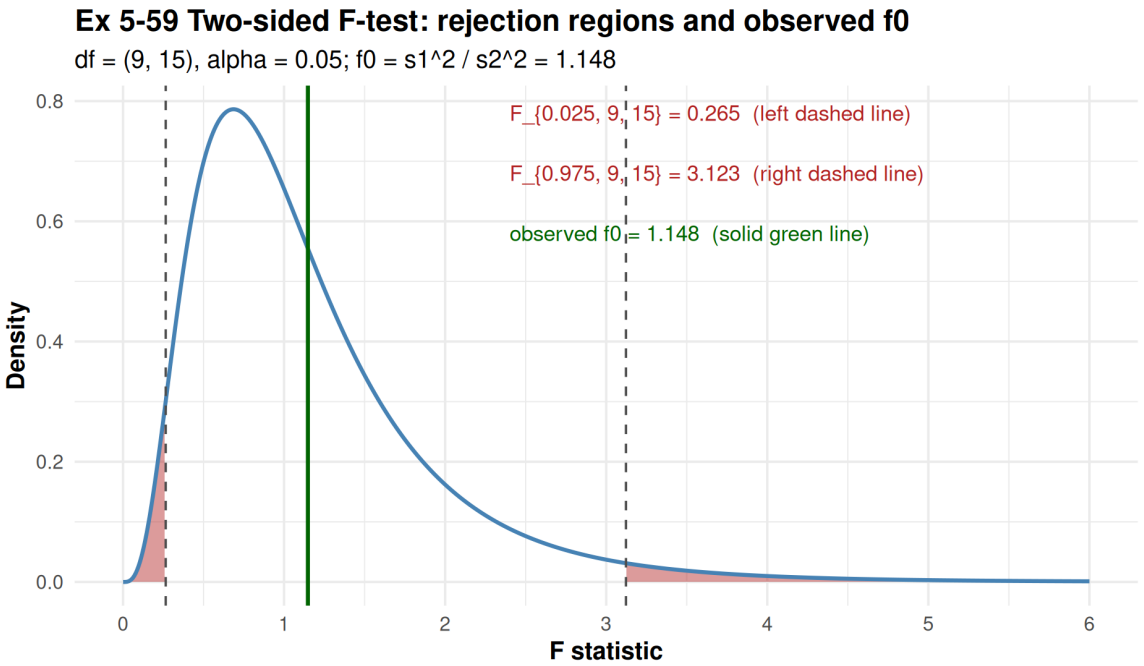

Figure 7: Ex 5-59 F(9,15) 分布、雙尾 $\alpha = 0.05$ 拒絕區（淺紅色）、觀察到的 $f_0 = 1.148$（綠色實線）。f_0 完全落在兩條虛線之間的接受區裡，所以不拒絕『兩變異數相等』。

最終答案 Final Answer

- **(a)** $f_0 = s_1^2/s_2^2 = 22.5^2/21^2 = 1.148$、$df = (9, 15)$；雙尾 P-value $= 0.781$，臨界值 $F_{0.025,9,15} = 0.265$、$F_{0.975,9,15} = 3.123$，f_0 落在拒絕區外 $\Rightarrow$ **FAIL TO REJECT** H_0。
- **(b)** 95% CI for $\sigma_1^2/\sigma_2^2 = (0.368,\ 4.327)$，1 在 CI 裡，跟 P-value 同一個結論。
- 解讀：兩 supplier 的衝擊強度變異數沒有顯著差異，這也支持了 5-24 用 Welch t（雖然不需要相等變異數）的選擇是穩妥的。

Exercise 5-69: Rollover Rates (Two Proportions)

5-69. The rollover rate of sport utility vehicles is a transportation safety issue. Safety advocates claim that manufacturer A's vehicle has a higher rollover rate than that of manufacturer B. One hundred crashes for each of these vehicles were examined. The rollover rates were $\hat{p}_A = 0.35$ and $\hat{p}_B = 0.25$.

(a) Does manufacturer A's vehicle have a higher rollover rate than manufacturer B's? Use the P-value approach.

(b) What is the power of this test, assuming $\alpha = 0.05$?

(c) Assume that the rollover rate of manufacturer A's vehicle is 0.15 higher than B's. Is the sample size sufficient for detecting this difference with probability level at least 0.90, if $\alpha = 0.05$?

助教校正：*(b)(c)* 內的 $p_A = 0.4$、$p_B = 0.25$ 是 H_1 下的真實 p，跟資料的 $\hat{p}_A$、$\hat{p}_B$ 是不同概念，不是改題目。

```r
# ==============================================================================
# EXERCISE 5-69: Vehicle Rollover Rates  (Section 5-6)
# Two proportions; n = c(100, 100); x = c(35, 25)
# 助教校正：(b)(c) 的 pA = 0.4, pB = 0.25 是 H1 下的 p (power / sample size 用)
# ==============================================================================
print_header("EXERCISE 5-69: ROLLOVER RATES (TWO PROPORTIONS)")

n     <- c(100, 100)
x     <- c(35, 25)
alpha <- 0.05

st <- TwoPropStats(x, n)

# --- (a) One-sided test: pA > pB ---
print_subhead("(a) One-sided two-proportion z-test (P-value approach)")
(z0    <- st$phat.diff / st$se.p)
(PVal  <- 1 - pnorm(z0))
cat("H0: pA  = pB\n")
cat("H1: pA  > pB\n")
cat(sprintf("phat.A   = %.3f , phat.B   = %.3f\n", st$phat[1], st$phat[2]))
cat(sprintf("phat.p   = %.3f , se.p     = %.3f\n", st$phat.p, st$se.p))
cat(sprintf("z0       = %.3f\n", z0))
cat(sprintf("P-value  = %.3f\n", PVal))
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# prop.test 對結果 (correct = FALSE 才能跟 z 公式對到)
pchk <- prop.test(x, n, alternative = "greater", correct = FALSE)
cat(sprintf("[prop.test]  chi^2 = %.3f, p = %.3f\n",
            pchk$statistic, pchk$p.value))

# --- (b) Power at alpha = 0.05, pA = 0.4, pB = 0.25 (H1 下) ---
print_subhead("(b) Power assuming pA = 0.4, pB = 0.25 (H1)")
pA_b <- 0.40
pB_b <- 0.25
pwr_b <- Pwr.prop2.test(n = n, pA = pA_b, pB = pB_b,
                        alpha = alpha, alt = "greater")
cat(sprintf("Under H1: pA = %.2f, pB = %.2f -> pA - pB = %.2f\n",
            pA_b, pB_b, pA_b - pB_b))
cat(sprintf("Power = %.3f , beta = %.3f\n", pwr_b, 1 - pwr_b))

# --- (c) 樣本數夠不夠 ---
print_subhead("(c) Is n = 100 each enough for power >= 0.90?")
n_req <- SplSz.prop2.test(pA = pA_b, pB = pB_b,
```

```r
                          alpha = alpha, pwr = 0.90, side = 1)
pwr_at100 <- Pwr.prop2.test(n = c(100, 100), pA = pA_b, pB = pB_b,
                            alpha = alpha, alt = "greater")
cat(sprintf("Required n (each group) = %d\n", n_req))
cat(sprintf("Power at current n = 100 each: %.3f\n", pwr_at100))
cat(" 結論：n=100 還沒到 0.9 power，按公式得補到 ", n_req, " 每組才夠。\n", sep = "")

# Power 對「每組樣本數」的關係圖：把現有 n=100 跟需要的 n=166 都標出來
n_grid <- seq(30, 250, by = 2)
pwr_grid_5_69 <- tibble(
  n_each = n_grid,
  power  = vapply(n_grid,
                  function(nn) Pwr.prop2.test(c(nn, nn),
                                              pA = pA_b, pB = pB_b,
                                              alpha = alpha, alt = "greater"),
                  numeric(1))
)
p_pwr_5_69 <- ggplot(pwr_grid_5_69, aes(n_each, power)) +
  geom_line(color = "steelblue", linewidth = 1.1) +
  geom_hline(yintercept = 0.90, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = 100,   linetype = "dotted",  color = "darkgreen") +
  geom_vline(xintercept = n_req, linetype = "dotted",  color = "purple") +
  geom_point(aes(x = 100,   y = pwr_at100), color = "darkgreen", size = 3) +
  geom_point(aes(x = n_req, y = 0.90),      color = "purple",    size = 3) +
  annotate("text", x = 100, y = pwr_at100 - 0.08,
           label = sprintf("n = 100,\npower = %.3f", pwr_at100),
           color = "darkgreen", size = 3.8, hjust = 1.05) +
  annotate("text", x = n_req, y = 0.82,
           label = sprintf("n = %d for\npower >= 0.90", n_req),
           color = "purple", size = 3.8, hjust = -0.05) +
  scale_y_continuous(breaks = seq(0, 1, 0.1), limits = c(0, 1.02)) +
  labs(title = "Ex 5-69 Two-proportion test: power vs per-group sample size",
       subtitle = "Under H1: pA = 0.40, pB = 0.25; alpha = 0.05; one-sided",
       x = "Per-group sample size  n",
       y = expression(paste("Power  = 1 - ", beta))) +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-69_power_vs_n.png", p_pwr_5_69,
       width = 8.0, height = 5.0, dpi = 220)
#>
#> ============================================================
#>  EXERCISE 5-69: ROLLOVER RATES (TWO PROPORTIONS)
#> ============================================================
#>
#> --- (a) One-sided two-proportion z-test (P-value approach) ---
#> [1] 1.543033
#> [1] 0.06141132
#> H0: pA  = pB
#> H1: pA  > pB
#> phat.A   = 0.350 , phat.B   = 0.250
#> phat.p   = 0.300 , se.p     = 0.065
#> z0       = 1.543
#> P-value  = 0.061
#> Decision:  FAIL TO REJECT H0
```

```
#> [prop.test]  chi^2 = 2.381, p = 0.061
#>
#> --- (b) Power assuming pA = 0.4, pB = 0.25 (H1) ---
#> Under H1: pA = 0.40, pB = 0.25 -> pA - pB = 0.15
#> Power = 0.735 , beta = 0.265
#>
#> --- (c) Is n = 100 each enough for power >= 0.90? ---
#> Required n (each group) = 166
#> Power at current n = 100 each: 0.735
#> 結論：n=100 還沒到 0.9 power，按公式得補到 166 每組才夠。
```

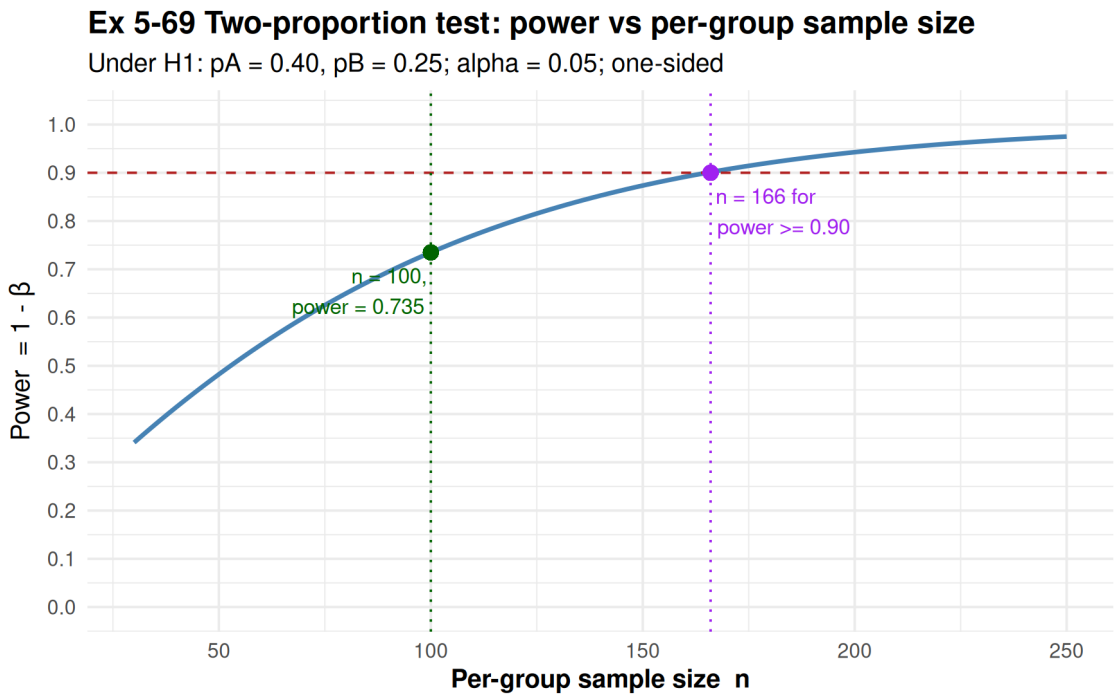

Figure 8: Ex 5-69 在 H_1：$p_A = 0.40, p_B = 0.25$ 下，power 對每組樣本數 n 的曲線。綠點 $n = 100$ 對應 power $= 0.735$、紫點是達到 power ≥ 0.90 的 $n = 166$；紅虛線是目標 power。

最終答案 Final Answer

- **(a)** pooled $\bar{p} = 0.300$、$SE_0 = 0.065$、$z_0 = 1.543$；單尾 P-value $= 0.061 > 0.05 \Rightarrow$ **FAIL TO REJECT** H_0：在 $\alpha = 0.05$ 下還不夠拒絕「A、B 翻覆率相同」。
- **(b)** 在 H_1 下 $p_A = 0.4$、$p_B = 0.25$、$n = (100, 100)$、$\alpha = 0.05$，power $= 0.735$、$\beta = 0.265$。
- **(c)** 需 power ≥ 0.90，由 $n = \lceil [z_{0.05}\sqrt{2\bar{p}\bar{q}} + z_{0.10}\sqrt{p_A q_A + p_B q_B}]^2/(p_A - p_B)^2 \rceil =$ **166** 每組。當前 $n = 100$ 不夠，power 只有 0.735。

Exercise 5-71: 95% Lower Bound on $p_A - p_B$

> **5-71.** Construct a 95% lower bound on the difference in the two rollover rates for Exercise 5-69. Provide a practical interpretation of this interval.

```r
# ==============================================================================
# EXERCISE 5-71: 95% Lower Bound on pA - pB  (Section 5-6)
# 沿用 5-69 資料；Wald 一邊下界
# ==============================================================================
print_header("EXERCISE 5-71: 95% LOWER CONFIDENCE BOUND FOR pA - pB")

n  <- c(100, 100)
x  <- c(35, 25)
alpha <- 0.05
st <- TwoPropStats(x, n)

LB <- st$phat.diff - qnorm(1 - alpha) * st$sehat
cat(sprintf("phat.diff = %.3f , sehat = %.3f\n", st$phat.diff, st$sehat))
cat(sprintf("z_{0.05}  = %.3f\n", qnorm(1 - alpha)))
cat(sprintf("Lower bound LB = phat.diff - z_{0.05} * sehat = %.3f\n", LB))
cat(sprintf("95%% one-sided lower bound : [ %.3f , 1 ]\n", LB))
# 解釋
if (LB > 0) {
  cat("LB > 0 -> 至少 95% 信心保證 pA > pB；和 5-69(a) 結論一致。\n")
} else {
  cat("LB <= 0 -> 0 還在區間裡，沒有 95% 信心斷言 pA > pB；跟 5-69(a) 邊界 P-value 一致。\n")
}
#>
#> ============================================================
#>  EXERCISE 5-71: 95% LOWER CONFIDENCE BOUND FOR pA - pB
#> ============================================================
#> phat.diff = 0.100 , sehat = 0.064
#> z_{0.05}  = 1.645
#> Lower bound LB = phat.diff - z_{0.05} * sehat = -0.006
#> 95% one-sided lower bound : [ -0.006 , 1 ]
#> LB <= 0 -> 0 還在區間裡，沒有 95% 信心斷言 pA > pB；跟 5-69(a) 邊界 P-value 一致。
```

最終答案 Final Answer

- 用傳統 Wald 形式：$\widehat{SE} = \sqrt{\hat{p}_A\hat{q}_A/n_A + \hat{p}_B\hat{q}_B/n_B} = 0.064$。
- $LB = (\hat{p}_A - \hat{p}_B) - z_{0.05}\widehat{SE} = 0.100 - 1.645 \cdot 0.064 = \mathbf{-0.006}$；95% one-sided lower bound 為 $[-0.006,\ 1]$。
- 解讀：下界微微落在 0 以下，跟 5-69(a) 的 P-value $= 0.061$ 完全一致，在 95% 信心下沒辦法斷言 $p_A > p_B$。換成百分點來看，這個下界相當於「A 比 B 最多高出 100%、最少甚至低 0.6 個百分點」。

Exercise 5-73: New CI (Plus-Four / Agresti-Caffo)

> **5-73.** Rework Exercise 5-71 using the new CI. Compare this interval to the traditional one.

```r
# ==============================================================================
# EXERCISE 5-73: Plus-Four (Agresti-Caffo) Lower Bound  (Section 5-6)
# 沿用 5-69 資料；ntilde = n + 2, xtilde = x + 1
# ==============================================================================
print_header("EXERCISE 5-73: NEW CI (PLUS-FOUR / AGRESTI-CAFFO)")

n  <- c(100, 100)
x  <- c(35, 25)
alpha <- 0.05
st <- TwoPropStats(x, n)

LB_new <- st$ptilde.diff - qnorm(1 - alpha) * st$setilde
# 對照 5-71 的傳統下界
LB_trad <- st$phat.diff - qnorm(1 - alpha) * st$sehat

cat(sprintf("ntilde = (%d, %d)\n", st$ntilde[1], st$ntilde[2]))
cat(sprintf("xtilde = (%d, %d)\n", st$xtilde[1], st$xtilde[2]))
cat(sprintf("ptilde     = (%.3f, %.3f)\n", st$ptilde[1], st$ptilde[2]))
cat(sprintf("ptilde.diff = %.3f , setilde = %.3f\n",
            st$ptilde.diff, st$setilde))
cat(sprintf("New lower bound LB_new  = %.3f\n", LB_new))
cat(sprintf("Traditional bound LB    = %.3f (from 5-71)\n", LB_trad))
cat(sprintf("Difference (new - trad) = %.3f\n", LB_new - LB_trad))
cat(" 解讀：plus-four 把 (x, n) 換成 (x+1, n+2)，等於各組多 1 顆成功 1 顆失敗，\n")
cat("       ptilde 更靠近 0.5，setilde 微幅變大，下界比傳統 Wald 略保守。\n")

# 把兩條單邊下界畫成水平線段，0 標一條垂直參考
lb_df <- tibble(
  method = factor(c("Traditional Wald (5-71)",
                    "Plus-four / Agresti-Caffo (5-73)"),
                  levels = c("Traditional Wald (5-71)",
                             "Plus-four / Agresti-Caffo (5-73)")),
  point  = c(st$phat.diff,  st$ptilde.diff),
  lb     = c(LB_trad,       LB_new)
)
p_lb <- ggplot(lb_df, aes(y = method)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "firebrick") +
  geom_segment(aes(x = lb, xend = point, yend = method),
               linewidth = 1.8, color = "steelblue") +
  geom_point(aes(x = point), size = 4, color = "navy") +
  geom_point(aes(x = lb),    size = 3.6, color = "darkorange3") +
  geom_text(aes(x = lb, label = sprintf("LB = %.3f", lb)),
            color = "darkorange3", vjust = -1.4, size = 4) +
  geom_text(aes(x = point, label = sprintf("point = %.3f", point)),
            color = "navy", vjust = -1.4, hjust = 0.5, size = 4) +
  # 左右各保留留白，避免 labels 被切掉
  scale_x_continuous(breaks = seq(-0.05, 0.15, 0.025),
                     limits = c(-0.03, 0.135),
                     expand = expansion(mult = 0.04)) +
  labs(title = "Ex 5-71 vs 5-73: 95% one-sided lower bound on pA - pB",
       subtitle = "Both bounds straddle 0, so neither can conclude pA > pB at 95%
        ↪ confidence",
```

```
       x = expression(paste(p[A], " - ", p[B], "  (percentage points)")),
       y = NULL) +
  theme_minimal(base_size = 13) +
  theme(plot.title  = element_text(face = "bold"),
        axis.title  = element_text(face = "bold"),
        axis.text.y = element_text(face = "bold", size = 11),
        plot.margin = margin(10, 18, 10, 10))
ggsave("figures/5-71-73_lb_compare.png", p_lb,
       width = 9.0, height = 4.2, dpi = 220)


# ==============================================================================
# 結束
# ==============================================================================
cat("\n============================================================\n")
cat(" All Chapter 5 exercises completed.\n")
cat("============================================================\n")
#>
#> ============================================================
#>  EXERCISE 5-73: NEW CI (PLUS-FOUR / AGRESTI-CAFFO)
#> ============================================================
#> ntilde = (102, 102)
#> xtilde = (36, 26)
#> ptilde     = (0.353, 0.255)
#> ptilde.diff = 0.098 , setilde = 0.064
#> New lower bound LB_new  = -0.007
#> Traditional bound LB    = -0.006 (from 5-71)
#> Difference (new - trad) = -0.001
#> 解讀：plus-four 把 (x, n) 換成 (x+1, n+2)，等於各組多 1 顆成功 1 顆失敗，
#>        ptilde 更靠近 0.5，setilde 微幅變大，下界比傳統 Wald 略保守。
#>
#> ============================================================
#>  All Chapter 5 exercises completed.
#> ============================================================
```

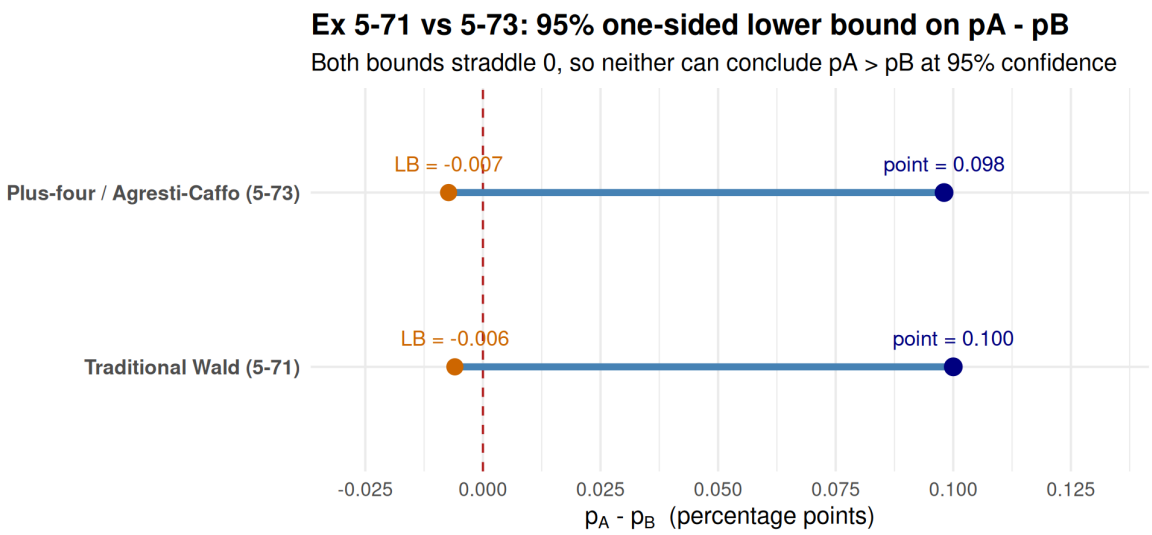

Figure 9: Ex 5-71 與 5-73 兩種 95% 單邊下界並列。橘點是下界、深藍點是點估計，紅虛線在 0。兩種方法的下界都落在 0 左側，所以都無法在 95% 信心下斷言 $p_A > p_B$。

最終答案 Final Answer

- 方法：$\tilde{n} = n + 2 = (102, 102)$、$\tilde{x} = x + 1 = (36, 26)$；$\tilde{p} = (0.353, 0.255)$、$\widetilde{SE} = 0.064$。
- $LB_{\text{new}} = (\tilde{p}_A - \tilde{p}_B) - z_{0.05}\widetilde{SE} = 0.098 - 1.645 \cdot 0.064 = \mathbf{-0.007}$。
- 比較：傳統 $LB = -0.006$、new $LB = -0.007$，相差 -0.001。plus-four 把 (x, n) 換成 $(x+1, n+2)$ 後點估計更靠近 0.5、$\widetilde{SE}$ 略大，所以下界比傳統 Wald 稍微保守一點點。
- 結論：兩種方法都跨過 0，定性結論一致：在 95% 信心下沒辦法斷言 $p_A > p_B$。Plus-four 的價值體現在 n 不大或 $\hat{p}$ 接近 $0/1$ 時的實際涵蓋率，本題差距其實非常小。

結語 / Closing Notes

整份作業跑完後，幾個容易踩到、但這次都有刻意守住的細節整理一下：

- 資料原樣保留：每題的 σ、n、$\bar{x}$、s、x 都照題目原文 hard-code，沒有臨時改數字。
- 單尾 / 雙尾要分清楚：5-3、5-5、5-17、5-41、5-59、5-73 是雙尾或 CI 形式；5-24、5-69、5-71 都是單尾。方向搞錯結論會反過來。
- 已知 σ **vs** 未知 σ：5-3、5-5、5-9 用 z（題目給 σ）；5-17、5-24、5-41 都用 t（資料給的是 s）。
- 變異數相等 **vs** 不等：5-17 題目明說相等 $\Rightarrow$ pooled t；5-24 題目明說不等 $\Rightarrow$ Welch t。5-59 用 F 檢定回頭驗證 supplier 1 / 2 的變異數其實沒有顯著差異，所以 5-24 用 Welch t 是穩妥的。
- 配對 **vs** 獨立樣本：5-41 是 12 個程式員在兩種語言下各跑一次，必須做 paired t。
- 比例的 **SE**：null hypothesis 用 pooled SE (5-69 (a))、CI 跟 power 用 unpooled SE (5-69 (b)、5-71、5-73)，這條分界一定要拿穩。
- 助教校正：5-69(b)(c) 的 $p_A = 0.4$、$p_B = 0.25$ 是 H_1 下假設的真實 p，不是把資料的 $\hat{p}$ 改掉。
- 樣本數一律 **`ceiling()`**：5-3、5-9、5-17、5-69 都用無條件進位，並印 $n-1$ 跟 n 兩個點的 power 來驗證。

R 4.5.2 跑完所有題目，繪圖用 `ggplot2` 與 `car::qqPlot`；套件交叉驗證用 `BSDA::z.test`、`BSDA::zsum.test`、`BSDA::tsum.test`、`stats::t.test`、`stats::prop.test`、`stats::power.t.test`。

附錄 A：完整 R 原始碼 (homework_assignments_2.R)

```
# ==============================================================================
# Homework Assignments 2  ----  Chapter 5 Decision Making for Two Samples
# Exercises: 5-3, 5-5, 5-9, 5-17, 5-24, 5-41, 5-59, 5-69, 5-71, 5-73
# ==============================================================================

## ---- init ----
# --- 1. Setup ------------------------------------------------------------
# 老師 R 範例都用 pacman 統一載套件，這邊照辦
if (!require(pacman)) install.packages("pacman", repos = "https://cloud.r-project.org")
pacman::p_load(HH, BSDA, ggplot2, purrr, dplyr, car, tibble, tidyr)

# 圖通通丟到 figures/，沒資料夾就先開
dir.create("figures", showWarnings = FALSE)

# 跟 console 一樣的標題分隔線，方便對版
print_header <- function(title) {
  cat("\n==============================================================\n")
  cat(" ", title, "\n", sep = "")
  cat("==============================================================\n")
}
print_subhead <- function(title) {
  cat(sprintf("\n--- %s ---\n", title))
}

# 全班預設四捨五入到第三位小數，這支腳本通用
fmt3 <- function(x) formatC(x, format = "f", digits = 3)


# --- 2. Reusable Functions ------------------------------------------------

# 兩樣本（含單樣本）的 z 檢定 power，這邊 sigma 可吃單一值或長度 2 向量
# delta 已經吃「真實平均差 - 虛無值」，所以呼叫端要自己算好
Pwr.z.test <- function(n, delta, sigma, alpha, alt = "two.sided") {
  # n 跟 sigma 都可以是長度 2 的向量；若只給一個就當兩組相同
  if (length(n) == 1)     n     <- rep(n,     2)
  if (length(sigma) == 1) sigma <- rep(sigma, 2)
  se <- sqrt(sum(sigma^2 / n))
  m  <- delta / se
  if (alt == "two.sided") {
    return(1 - pnorm(qnorm(1 - alpha/2) - m) + pnorm(qnorm(alpha/2) - m))
  } else if (alt == "greater") {
    return(1 - pnorm(qnorm(1 - alpha) - m))
  } else {
    return(pnorm(qnorm(alpha) - m))
  }
}

# 兩樣本 z 檢定的「等樣本數」公式：sigma 可吃單一或向量
# 課本第 5-2 節公式：n = (z_{a/side} + z_{pwr})^2 * (sigma1^2 + sigma2^2) / delta^2
SplSz.z.test <- function(delta, sigma, alpha, pwr, side = 2) {
  if (length(sigma) == 1) sigma <- rep(sigma, 2)
  ceiling(((qnorm(1 - alpha/side) + qnorm(pwr))^2 * sum(sigma^2)) / delta^2)
}

# 兩樣本 z 區間估計的等樣本數：CI 寬要 <= 2E
# n = (z_{(1-conf)/side} * sqrt(sigma1^2 + sigma2^2) / E)^2
SplSz.z.CI <- function(E, sigma, conf.level, side = 2) {
  if (length(sigma) == 1) sigma <- rep(sigma, 2)
  ceiling((qnorm(1 - (1 - conf.level)/side))^2 * sum(sigma^2) / E^2)
```

```
}

# Pooled t 的 power 直接走 stats::power.t.test，介面跟老師範例一致
# delta 是真平均差，sigma 是 pooled SD 的估計值
Pwr.t.test <- function(n, delta, sigma, alpha, alt = "two.sided") {
  side <- if (alt == "two.sided") "two.sided" else "one.sided"
  power.t.test(n         = n,
               delta     = abs(delta),
               sd        = sigma,
               sig.level = alpha,
               type      = "two.sample",
               alternative = side)$power
}

# 兩比例的標準誤、調整估計值（Agresti-Caffo plus-four）一次包好，呼叫端拿去組
TwoPropStats <- function(x, n) {
  phat       <- x / n
  qhat       <- 1 - phat
  phat.diff  <- -diff(phat)               # p1 - p2
  phat.p     <- sum(x) / sum(n)           # pooled point estimate
  qhat.p     <- 1 - phat.p
  se.p       <- sqrt(phat.p * qhat.p * sum(1 / n))   # pooled SE (for null test)
  sehat      <- sqrt(sum(phat * qhat / n))           # unpooled SE (for CI)
  # plus-four / Agresti-Caffo
  ntilde     <- n + 2
  xtilde     <- x + 1
  ptilde     <- xtilde / ntilde
  qtilde     <- 1 - ptilde
  ptilde.diff <- -diff(ptilde)
  setilde    <- sqrt(sum(ptilde * qtilde / ntilde))
  list(phat = phat, qhat = qhat, phat.diff = phat.diff,
       phat.p = phat.p, qhat.p = qhat.p, se.p = se.p, sehat = sehat,
       ntilde = ntilde, xtilde = xtilde, ptilde = ptilde,
       qtilde = qtilde, ptilde.diff = ptilde.diff, setilde = setilde)
}

# 兩比例檢定的 power（課本第 5-6 節公式，已知 pA、pB）
# 拒絕域用 pooled SE，power 在 H1 下用 unpooled SE
Pwr.prop2.test <- function(n, pA, pB, alpha, alt = "greater") {
  if (length(n) == 1) n <- rep(n, 2)
  p_bar  <- (n[1]*pA + n[2]*pB) / sum(n)
  q_bar  <- 1 - p_bar
  se0    <- sqrt(p_bar * q_bar * sum(1 / n))
  se1    <- sqrt(pA*(1-pA)/n[1] + pB*(1-pB)/n[2])
  d      <- pA - pB
  if (alt == "greater") {
    return(1 - pnorm((qnorm(1 - alpha) * se0 - d) / se1))
  } else if (alt == "less") {
    return(pnorm((qnorm(alpha) * se0 - d) / se1))
  } else {
    return(1 - pnorm(( qnorm(1 - alpha/2) * se0 - d) / se1)
           +     pnorm((-qnorm(1 - alpha/2) * se0 - d) / se1))
  }
}

# 兩比例的「等樣本數」近似公式
SplSz.prop2.test <- function(pA, pB, alpha, pwr, side = 2) {
  d <- pA - pB
  p_bar <- (pA + pB) / 2
  q_bar <- 1 - p_bar
  z_a <- qnorm(1 - alpha/side)
```

```
  z_b <- qnorm(pwr)
  num <- z_a * sqrt(2 * p_bar * q_bar) + z_b * sqrt(pA*(1-pA) + pB*(1-pB))
  ceiling((num / d)^2)
}

# 描述統計小工具，順便壓三位小數方便看
get_stats <- function(x) {
  data.frame(
    n         = length(x),
    Mean      = mean(x),
    Var       = var(x),
    Std_Dev   = sd(x),
    Std_Error = sd(x) / sqrt(length(x)),
    Median    = median(x)
  ) %>% mutate(across(everything(), ~ round(., 3)))
}

# Q-Q + 直方圖一次存（給 5-41 paired diff 用）
# x_label 走具體的軸名，避免印出 "Value" 這種沒資訊量的字
save_normality_plots <- function(x, title_tag, file_prefix, x_label = "Value") {
  df <- data.frame(value = x)
  sw <- shapiro.test(x)
  sw_sub <- sprintf("Shapiro-Wilk: W = %.3f, p = %.3f", sw$statistic, sw$p.value)
  stats_sub <- sprintf("n = %d, x_bar = %.3f, s = %.3f",
                       length(x), mean(x), sd(x))
  p_qq <- ggplot(df, aes(sample = value)) +
    stat_qq(color = "steelblue", size = 2.4) +
    stat_qq_line(color = "firebrick", linewidth = 0.9) +
    labs(title = paste("Normal Q-Q Plot:", title_tag),
         subtitle = sw_sub,
         x = "Theoretical normal quantiles",
         y = paste0("Sample quantiles of ", x_label)) +
    theme_minimal(base_size = 13) +
    theme(plot.title    = element_text(face = "bold"),
          axis.title    = element_text(face = "bold"))
  bins_used <- if (length(x) <= 6) 4 else max(5, min(10, ceiling(log2(length(x)) + 1)))
  p_hist <- ggplot(df, aes(x = value)) +
    geom_histogram(aes(y = after_stat(density)),
                   bins = bins_used,
                   fill = "steelblue", color = "white", alpha = 0.85) +
    geom_rug(color = "navyblue", sides = "b", length = unit(0.04, "npc")) +
    stat_function(fun = dnorm,
                  args = list(mean = mean(x), sd = sd(x)),
                  color = "firebrick", linewidth = 1) +
    labs(title = paste("Histogram with Normal Curve:", title_tag),
         subtitle = stats_sub,
         x = x_label, y = "Density") +
    theme_minimal(base_size = 13) +
    theme(plot.title    = element_text(face = "bold"),
          axis.title    = element_text(face = "bold"))
  ggsave(paste0("figures/", file_prefix, "_qq.png"),
         p_qq, width = 7.5, height = 5.2, dpi = 220)
  ggsave(paste0("figures/", file_prefix, "_hist.png"),
         p_hist, width = 7.5, height = 5.2, dpi = 220)
}


## ---- ex5_03 ----
# ==============================================================================
# EXERCISE 5-3: Bottle Filling Machines  (Section 5-2)
# 兩台機台、已知 sigma1 = 0.020, sigma2 = 0.025；雙尾、alpha = 0.05
```

```r
# ==============================================================================
print_header("EXERCISE 5-3: BOTTLE FILLING MACHINES")

# 題目資料原樣 hard-code，每組 10 筆
Machine1 <- c(16.03, 16.01,
              16.04, 15.96,
              16.05, 15.98,
              16.05, 16.02,
              16.02, 15.99)

Machine2 <- c(16.02, 16.03,
              15.97, 16.04,
              15.96, 16.02,
              16.01, 16.01,
              15.99, 16.00)

params <- list(sigma  = c(0.020, 0.025),
               n      = c(length(Machine1), length(Machine2)),
               Delta0 = 0,
               alpha  = 0.05,
               conf.level = 0.95)

# --- 1. Basic Stats ---
print_subhead("Sample summary")
print(get_stats(Machine1))
print(get_stats(Machine2))

# 老師常用括號賦值，順手把 xbar、se0、z0 印出來
(xbar  <- c(mean(Machine1), mean(Machine2)))
(se0   <- sqrt(sum(params$sigma^2 / params$n)))
(z0    <- (xbar[1] - xbar[2] - params$Delta0) / se0)

# --- 2. (a) P-value 雙尾 ---
print_subhead("(a) Two-sided z-test (known variances)")
PVal <- 2 * (1 - pnorm(abs(z0)))
cat("H0: mu1  = mu2\n")
cat("H1: mu1 != mu2\n")
cat(sprintf("xbar     = (%.3f, %.3f)\n", xbar[1], xbar[2]))
cat(sprintf("xbar.diff = %.3f\n", xbar[1] - xbar[2]))
cat(sprintf("se0      = %.3f\n",  se0))
cat(sprintf("z0       = %.3f\n",  z0))
cat(sprintf("P-value  = %.3f\n",  PVal))
cat("Decision: ",
    if (PVal < params$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")

# 套件交叉驗證：BSDA::z.test 走的是同一條路
zchk <- BSDA::z.test(Machine1, Machine2,
                     sigma.x = params$sigma[1],
                     sigma.y = params$sigma[2],
                     alternative = "two.sided",
                     conf.level  = params$conf.level)
cat(sprintf("[BSDA::z.test]  z = %.3f, p = %.3f\n",
            zchk$statistic, zchk$p.value))

# --- 3. (b) Power for true diff = 0.04 ---
print_subhead("(b) Power for true |mu1 - mu2| = 0.04 (alpha = 0.05)")
delta_b <- 0.04
pwr_b   <- Pwr.z.test(n = params$n, delta = delta_b,
                      sigma = params$sigma, alpha = params$alpha,
                      alt = "two.sided")
cat(sprintf("delta    = %.3f\n", delta_b))
```

```
cat(sprintf("Power    = %.3f\n", pwr_b))
cat(sprintf("Beta     = %.3f\n", 1 - pwr_b))

# --- 4. (c) 95% CI for mu1 - mu2 ---
print_subhead("(c) 95% Confidence Interval for mu1 - mu2")
(CritVal <- qnorm(1 - params$alpha/2))
CI <- c((xbar[1] - xbar[2]) - CritVal * se0,
        (xbar[1] - xbar[2]) + CritVal * se0)
cat(sprintf("CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[BSDA::z.test CI] (%.3f , %.3f)\n",
            zchk$conf.int[1], zchk$conf.int[2]))

# --- 5. (d) Sample size for beta = 0.01, alpha = 0.05, delta = 0.04 ---
print_subhead("(d) Equal n so that beta = 0.01 when |mu1 - mu2| = 0.04")
n_req <- SplSz.z.test(delta = 0.04, sigma = params$sigma,
                      alpha = params$alpha, pwr = 0.99, side = 2)
# n_req - 1 跟 n_req 兩個點的 power 都印，方便驗證
pw_m1 <- Pwr.z.test(n_req - 1, delta = 0.04, sigma = params$sigma,
                    alpha = params$alpha, alt = "two.sided")
pw_n  <- Pwr.z.test(n_req,     delta = 0.04, sigma = params$sigma,
                    alpha = params$alpha, alt = "two.sided")
cat(sprintf("Required n (each group) = %d\n", n_req))
cat(sprintf("Power at n-1 (%d) = %.3f , beta = %.3f\n",
            n_req - 1, pw_m1, 1 - pw_m1))
cat(sprintf("Power at n   (%d) = %.3f , beta = %.3f\n",
            n_req,     pw_n,  1 - pw_n))

# Power curve：以 delta 為 x 軸
pwr_grid <- tibble(
  delta = seq(0, 0.10, length.out = 200)
) %>% mutate(
  power = Pwr.z.test(params$n, delta, params$sigma, params$alpha, "two.sided")
)
p_pwr_5_3 <- ggplot(pwr_grid, aes(delta, power)) +
  geom_line(color = "steelblue", linewidth = 1.1) +
  geom_hline(yintercept = 0.99, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = 0.04, linetype = "dotted",  color = "darkgreen") +
  annotate("text", x = 0.042, y = 0.45, label = "delta = 0.04",
           color = "darkgreen", hjust = 0, size = 4) +
  annotate("text", x = 0.08, y = 0.965, label = "target power = 0.99",
           color = "firebrick", hjust = 0, size = 4) +
  scale_y_continuous(breaks = seq(0, 1, 0.25), limits = c(0, 1.02)) +
  labs(title = "Ex 5-3 Two-sample z-test power curve",
       subtitle = "known sigma1 = 0.020, sigma2 = 0.025; n1 = n2 = 10; alpha = 0.05",
       x = expression(paste("True difference  |", mu[1] - mu[2], "|  (oz)")),
       y = expression(paste("Power  = 1 - ", beta))) +
  theme_minimal(base_size = 13) +
  theme(plot.title  = element_text(face = "bold"),
        axis.title  = element_text(face = "bold"))
ggsave("figures/5-3_power_curve.png", p_pwr_5_3,
       width = 7.5, height = 5.0, dpi = 220)


## ---- ex5_05 ----
# ==============================================================================
# EXERCISE 5-5: Solid-Fuel Propellants  (Section 5-2)
# 已知 sigma1 = sigma2 = 3 cm/s; n1 = n2 = 20; xbar1 = 18.02, xbar2 = 24.37
# ==============================================================================
print_header("EXERCISE 5-5: SOLID-FUEL PROPELLANT BURNING RATES")

# 只有 summary 給；不過 BSDA::zsum.test 剛好吃 summary
```

```r
xbar  <- c(18.02, 24.37)
sigma <- c(3, 3)
n     <- c(20, 20)
alpha <- 0.05

(se0 <- sqrt(sum(sigma^2 / n)))
(z0  <- (xbar[1] - xbar[2]) / se0)

# --- (a)(b) fixed-level + P-value ---
print_subhead("(a)(b) Two-sided z-test, alpha = 0.05")
(CritVal <- qnorm(1 - alpha/2))
PVal <- 2 * (1 - pnorm(abs(z0)))
cat("H0: mu1  = mu2\n")
cat("H1: mu1 != mu2\n")
cat(sprintf("se0     = %.3f\n", se0))
cat(sprintf("z0      = %.3f\n", z0))
cat(sprintf("CritVal = +/- %.3f\n", CritVal))
cat(sprintf("P-value = %.3f\n", PVal))
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# BSDA::zsum.test 提供同樣的結果，可比對
zsum_chk <- BSDA::zsum.test(mean.x = xbar[1], sigma.x = sigma[1], n.x = n[1],
                            mean.y = xbar[2], sigma.y = sigma[2], n.y = n[2],
                            alternative = "two.sided", conf.level = 1 - alpha)
cat(sprintf("[BSDA::zsum.test]  z = %.3f, p = %.3f\n",
            zsum_chk$statistic, zsum_chk$p.value))

# --- (c) beta when true diff = 2.5 cm/s ---
print_subhead("(c) Beta when true mu1 - mu2 = 2.5 cm/s")
delta_c <- 2.5
pwr_c   <- Pwr.z.test(n = n, delta = delta_c, sigma = sigma,
                      alpha = alpha, alt = "two.sided")
cat(sprintf("delta = %.3f\n",  delta_c))
cat(sprintf("Power = %.3f\n",  pwr_c))
cat(sprintf("Beta  = %.3f\n",  1 - pwr_c))

# --- (d) 95% CI for mu1 - mu2 ---
print_subhead("(d) 95% CI for mu1 - mu2")
CI <- c((xbar[1] - xbar[2]) - CritVal * se0,
        (xbar[1] - xbar[2]) + CritVal * se0)
cat(sprintf("CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[zsum.test CI]   (%.3f , %.3f)\n",
            zsum_chk$conf.int[1], zsum_chk$conf.int[2]))

# Beta 區域圖：H0 跟 H1 兩條常態，beta 對應沒被拒絕的部分
# 拿真實 delta = 2.5 的設定畫，能看出 power=0.75 的幾何意義
beta_grid <- seq(-3, 6, length.out = 600)
beta_df <- tibble(
  d        = beta_grid,
  null     = dnorm(beta_grid, mean = 0,       sd = se0),
  alt      = dnorm(beta_grid, mean = 2.5,     sd = se0)
)
crit_lo <- -CritVal * se0
crit_hi <-  CritVal * se0
# 在 H1 下沒被拒絕的區間就是 beta
shade_df <- beta_df %>%
  filter(d >= crit_lo, d <= crit_hi) %>%
  mutate(ymin = 0)
p_beta_5_5 <- ggplot(beta_df, aes(d)) +
  geom_area(aes(y = alt), data = shade_df,
            fill = "orange", alpha = 0.45) +
```

```
  geom_line(aes(y = null), color = "steelblue", linewidth = 1) +
  geom_line(aes(y = alt),  color = "firebrick", linewidth = 1) +
  geom_vline(xintercept = c(crit_lo, crit_hi),
             linetype = "dashed", color = "gray30") +
  annotate("text", x = 0,   y = 0.5, label = "H0: mu1 - mu2 = 0",
           color = "steelblue", size = 4, hjust = 0.5) +
  annotate("text", x = 2.5, y = 0.5, label = "H1: mu1 - mu2 = 2.5",
           color = "firebrick", size = 4, hjust = 0.5) +
  annotate("text", x = 1.0, y = 0.12,
           label = sprintf("beta region\n(area = %.3f)", 1 - pwr_c),
           color = "darkorange3", size = 3.8, hjust = 0.5) +
  labs(title = "Ex 5-5 Two-sample z-test: beta region at true delta = 2.5",
       subtitle = "sigma = 3 cm/s, n1 = n2 = 20, alpha = 0.05; two-sided",
       x = expression(paste("Sample mean difference  ",
                            bar(x)[1] - bar(x)[2], "  (cm/s)")),
       y = "Density") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-5_beta_region.png", p_beta_5_5,
       width = 7.8, height = 5.0, dpi = 220)


## ---- ex5_09 ----
# ==============================================================================
# EXERCISE 5-9: Sample size for estimating mu1 - mu2  (Section 5-2)
# 沿用 5-5：sigma1 = sigma2 = 3; equal n; E = 4 cm/s; 99% confidence
# ==============================================================================
print_header("EXERCISE 5-9: SAMPLE SIZE FOR DIFFERENCE IN MEANS")

E <- 4
sigma <- c(3, 3)
conf.level <- 0.99
n_each <- SplSz.z.CI(E = E, sigma = sigma, conf.level = conf.level, side = 2)

# 跟手算公式對一下：n = ((z_{0.005})^2 * (sigma1^2 + sigma2^2)) / E^2
z_half <- qnorm(1 - (1 - conf.level)/2)
manual <- (z_half^2 * sum(sigma^2)) / E^2
cat(sprintf("Confidence level = %.3f\n", conf.level))
cat(sprintf("Margin of error E = %.0f cm/s\n", E))
cat(sprintf("sigma = (%.0f, %.0f)\n", sigma[1], sigma[2]))
cat(sprintf("z_{0.005} = %.3f\n", z_half))
cat(sprintf("Formula value = %.3f -> ceiling = %d (per group)\n",
            manual, n_each))


## ---- ex5_17 ----
# ==============================================================================
# EXERCISE 5-17: Single vs Dual Spindle Saw  (Section 5-3)
# Pooled t (假設變異數相同); n1 = n2 = 15
# xbar_s = 66.385, s_s = 7.895;  xbar_d = 45.278, s_d = 8.612
# ==============================================================================
print_header("EXERCISE 5-17: SINGLE vs DUAL SPINDLE SAW")

n     <- c(15, 15)
xbar  <- c(66.385, 45.278)   # single, double
s     <- c(7.895, 8.612)
alpha <- 0.05

# Pooled SD 跟自由度，老師範例都會印
(s_p <- sqrt(((n[1]-1)*s[1]^2 + (n[2]-1)*s[2]^2) / (sum(n) - 2)))
```

```r
(se  <- s_p * sqrt(sum(1/n)))
(t0  <- (xbar[1] - xbar[2]) / se)
df  <- sum(n) - 2

# --- (a) Two-sided P-value ---
print_subhead("(a) Two-sided pooled t-test")
PVal <- 2 * (1 - pt(abs(t0), df))
cat("H0: mu_single  = mu_double\n")
cat("H1: mu_single != mu_double\n")
cat(sprintf("s_p     = %.3f\n", s_p))
cat(sprintf("se      = %.3f\n", se))
cat(sprintf("t0      = %.3f\n", t0))
cat(sprintf("df      = %d\n",   df))
cat(sprintf("P-value = %.3g\n", PVal))   # 太小就用科學記號
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# 拿 BSDA::tsum.test 對結果
tsum_chk <- BSDA::tsum.test(mean.x = xbar[1], s.x = s[1], n.x = n[1],
                            mean.y = xbar[2], s.y = s[2], n.y = n[2],
                            var.equal = TRUE, alternative = "two.sided",
                            conf.level = 1 - alpha)
cat(sprintf("[BSDA::tsum.test]  t = %.3f, df = %.0f, p = %.3g\n",
            tsum_chk$statistic, tsum_chk$parameter, tsum_chk$p.value))

# --- (b) 95% two-sided CI ---
print_subhead("(b) 95% CI for mu_single - mu_double")
CritVal <- qt(1 - alpha/2, df)
CI <- c((xbar[1] - xbar[2]) - CritVal * se,
        (xbar[1] - xbar[2]) + CritVal * se)
cat(sprintf("t_{0.025, %d} = %.3f\n", df, CritVal))
cat(sprintf("CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[tsum.test CI]   (%.3f , %.3f)\n",
            tsum_chk$conf.int[1], tsum_chk$conf.int[2]))

# --- (c) Sample size for beta <= 0.10 when delta = 15 ---
print_subhead("(c) Equal n so that beta <= 0.10 when |mu1 - mu2| = 15")
# 拿 pooled s_p 當 sigma 估計，alpha=0.05，目標 power=0.9，雙尾
# power.t.test() 給 n 是「每組」樣本數，不用再除 2
spl_5_17 <- power.t.test(delta = 15, sd = s_p, sig.level = alpha,
                         power = 0.90, type = "two.sample",
                         alternative = "two.sided")
n_req <- ceiling(spl_5_17$n)
pw_m1 <- Pwr.t.test(n_req - 1, delta = 15, sigma = s_p, alpha = alpha)
pw_n  <- Pwr.t.test(n_req,     delta = 15, sigma = s_p, alpha = alpha)
cat(sprintf("sigma estimate (= s_p) = %.3f\n", s_p))
cat(sprintf("power.t.test raw n     = %.3f -> ceiling = %d (per group)\n",
            spl_5_17$n, n_req))
cat(sprintf("Power at n-1 (%d) = %.3f , beta = %.3f\n",
            n_req - 1, pw_m1, 1 - pw_m1))
cat(sprintf("Power at n   (%d) = %.3f , beta = %.3f\n",
            n_req,     pw_n,  1 - pw_n))

# t 分布拒絕區圖：df=28, 雙尾 alpha=0.05，順手把觀察到的 t0 也標出來
t_grid <- seq(-8, 8, length.out = 700)
t_df_plot <- tibble(t = t_grid, dens = dt(t_grid, df))
crit_t <- qt(1 - alpha/2, df)
reject_lo <- t_df_plot %>% filter(t <= -crit_t)
reject_hi <- t_df_plot %>% filter(t >=  crit_t)
p_t_5_17 <- ggplot(t_df_plot, aes(t, dens)) +
  geom_area(data = reject_lo, fill = "firebrick", alpha = 0.45) +
  geom_area(data = reject_hi, fill = "firebrick", alpha = 0.45) +
```

```
  geom_line(color = "steelblue", linewidth = 1) +
  geom_vline(xintercept = c(-crit_t, crit_t),
             linetype = "dashed", color = "gray30") +
  geom_vline(xintercept = t0, linetype = "solid",
             color = "darkgreen", linewidth = 1) +
  annotate("text", x = -crit_t, y = 0.35,
           label = sprintf("-t_{0.025, 28}\n= %.3f", -crit_t),
           color = "firebrick", size = 3.8, hjust = 1.05) +
  annotate("text", x = crit_t,  y = 0.35,
           label = sprintf("t_{0.025, 28}\n= %.3f", crit_t),
           color = "firebrick", size = 3.8, hjust = -0.05) +
  annotate("text", x = t0, y = 0.18,
           label = sprintf("observed t0\n= %.3f", t0),
           color = "darkgreen", size = 3.8, hjust = -0.05) +
  labs(title = "Ex 5-17 Pooled t-test: rejection regions and observed t0",
       subtitle = "df = 28, two-sided alpha = 0.05",
       x = "t statistic", y = "Density") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-17_t_rejection.png", p_t_5_17,
       width = 7.8, height = 5.0, dpi = 220)


## ---- ex5_24 ----
# ==============================================================================
# EXERCISE 5-24: Plastic Gear Impact Strength  (Section 5-3)
# Welch t (變異數不等); n1 = 10, xbar1 = 289.30, s1 = 22.5
#                       n2 = 16, xbar2 = 321.50, s2 = 21
# ==============================================================================
print_header("EXERCISE 5-24: PLASTIC GEAR SUPPLIERS")

n      <- c(10, 16)
xbar   <- c(289.30, 321.50)   # supplier 1, supplier 2
s      <- c(22.5, 21)
alpha  <- 0.05

# Welch SE 跟 Welch-Satterthwaite df
(se_W <- sqrt(s[1]^2/n[1] + s[2]^2/n[2]))
df_W  <- (s[1]^2/n[1] + s[2]^2/n[2])^2 /
         ((s[1]^2/n[1])^2 / (n[1]-1) + (s[2]^2/n[2])^2 / (n[2]-1))

# --- (a) one-sided test: supplier 2 higher than supplier 1 ---
print_subhead("(a) One-sided Welch t-test, H1: mu2 > mu1")
# 直接寫成 mu2 - mu1 - 0 比較好讀
t0_a <- ((xbar[2] - xbar[1]) - 0) / se_W
PVal_a <- 1 - pt(t0_a, df_W)
cat("H0: mu1 = mu2  (delta = mu2 - mu1 = 0)\n")
cat("H1: mu2 > mu1  (delta > 0)\n")
cat(sprintf("se (Welch)   = %.3f\n", se_W))
cat(sprintf("df (Welch)   = %.3f\n", df_W))
cat(sprintf("t0           = %.3f\n", t0_a))
cat(sprintf("P-value      = %.3f\n", PVal_a))
cat("Decision: ",
    if (PVal_a < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# tsum.test 對一遍
tsum_a <- BSDA::tsum.test(mean.x = xbar[2], s.x = s[2], n.x = n[2],
                          mean.y = xbar[1], s.y = s[1], n.y = n[1],
                          var.equal = FALSE,
                          alternative = "greater", conf.level = 1 - alpha)
cat(sprintf("[BSDA::tsum.test]  t = %.3f, df = %.3f, p = %.3f\n",
```

```
            tsum_a$statistic, tsum_a$parameter, tsum_a$p.value))

# --- (b) Test mu2 - mu1 >= 25 ---
print_subhead("(b) Test claim mu2 - mu1 >= 25 (Welch t)")
# 主張要靠資料「站得住腳」，就把它放在 H1
Delta0 <- 25
t0_b <- ((xbar[2] - xbar[1]) - Delta0) / se_W
PVal_b <- 1 - pt(t0_b, df_W)
cat("H0: mu2 - mu1  = 25\n")
cat("H1: mu2 - mu1  > 25\n")
cat(sprintf("observed diff = %.3f\n", xbar[2] - xbar[1]))
cat(sprintf("t0           = %.3f\n", t0_b))
cat(sprintf("P-value      = %.3f\n", PVal_b))
cat("Decision: ",
    if (PVal_b < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat("Interpretation: 沒有夠強的證據說『supplier 2 比 supplier 1 平均多 25 foot-pounds 以上』。\n")

# --- (c) 95% CI for mu2 - mu1 ---
print_subhead("(c) 95% CI for mu2 - mu1")
CritVal <- qt(1 - alpha/2, df_W)
CI <- c((xbar[2] - xbar[1]) - CritVal * se_W,
        (xbar[2] - xbar[1]) + CritVal * se_W)
cat(sprintf("t_{0.025, %.2f} = %.3f\n", df_W, CritVal))
cat(sprintf("CI for mu2 - mu1 = (%.3f , %.3f)\n", CI[1], CI[2]))
# 用 CI 回答 (b)：25 在區間裡 -> 不拒絕
cat(sprintf("Is 25 inside the CI? %s -> %s\n",
            ifelse(CI[1] <= 25 && 25 <= CI[2], "Yes", "No"),
            ifelse(CI[1] <= 25 && 25 <= CI[2],
                   "consistent with (b): FAIL TO REJECT",
                   "would REJECT H0 in (b)")))

# CI 圖：把點估計 + 95% CI + 兩條參考線 (0 與 25) 畫在同一條 X 軸上
ci_df_5_24 <- tibble(
  label = "mu2 - mu1",
  est   = xbar[2] - xbar[1],
  lo    = CI[1],
  hi    = CI[2]
)
p_ci_5_24 <- ggplot(ci_df_5_24, aes(y = label)) +
  geom_vline(xintercept = 0,  linetype = "dashed", color = "gray40") +
  geom_vline(xintercept = 25, linetype = "dotdash", color = "firebrick",
             linewidth = 0.8) +
  geom_segment(aes(x = lo, xend = hi, yend = label),
               linewidth = 2, color = "steelblue") +
  geom_point(aes(x = est), size = 4.5, color = "navy") +
  geom_text(aes(x = est, label = sprintf("%.3f", est)),
            vjust = -1.2, size = 4, color = "navy") +
  geom_text(aes(x = lo, label = sprintf("%.3f", lo)),
            vjust = 2.2, size = 3.6, color = "gray30") +
  geom_text(aes(x = hi, label = sprintf("%.3f", hi)),
            vjust = 2.2, size = 3.6, color = "gray30") +
  annotate("text", x = 0,  y = 0.6, label = "0 (test in (a))",
           color = "gray30",   size = 3.6, hjust = -0.05) +
  annotate("text", x = 25, y = 0.6, label = "25 (test in (b))",
           color = "firebrick", size = 3.6, hjust = -0.05) +
  scale_x_continuous(breaks = c(0, 10, 20, 25, 30, 40, 50)) +
  labs(title = "Ex 5-24 Welch 95% CI for mu2 - mu1, with H0 boundaries",
       subtitle = "n1 = 10, n2 = 16; Welch df = 18.226; SE = 8.842",
       x = "Difference in mean impact strength (foot-pounds)", y = NULL) +
  theme_minimal(base_size = 13) +
  theme(plot.title  = element_text(face = "bold"),
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        axis.title  = element_text(face = "bold"),
        axis.text.y = element_text(face = "bold", size = 12))
ggsave("figures/5-24_ci_forest.png", p_ci_5_24,
       width = 8.5, height = 4.2, dpi = 220)


## ---- ex5_41 ----
# ==============================================================================
# EXERCISE 5-41: Coding Time, Two Design Languages  (Section 5-4)
# 12 個程式員、配對 t；Coding.diff = Lang1 - Lang2
# ==============================================================================
print_header("EXERCISE 5-41: PAIRED CODING TIMES (TWO DESIGN LANGUAGES)")

Lang1 <- c(17, 16, 21, 14, 18, 24, 16, 14, 21, 23, 13, 18)
Lang2 <- c(18, 14, 19, 11, 23, 21, 10, 13, 19, 24, 15, 20)

# 方向：取 Lang1 - Lang2，表示「語言 1 比語言 2 多用多少分鐘」
Coding.diff <- Lang1 - Lang2
n_p <- length(Coding.diff)
df_p <- n_p - 1
alpha <- 0.05

print_subhead("Summary of paired differences (Lang1 - Lang2)")
print(get_stats(Coding.diff))

# --- (a) 95% CI on mean paired difference ---
print_subhead("(a) 95% CI on mean(Lang1 - Lang2)")
t_paired <- t.test(Lang1, Lang2, paired = TRUE, conf.level = 1 - alpha)
xbar_d <- mean(Coding.diff)
sd_d   <- sd(Coding.diff)
se_d   <- sd_d / sqrt(n_p)
CritVal <- qt(1 - alpha/2, df_p)
CI <- c(xbar_d - CritVal * se_d, xbar_d + CritVal * se_d)
cat(sprintf("xbar_d (Lang1 - Lang2) = %.3f\n", xbar_d))
cat(sprintf("s_d                    = %.3f\n", sd_d))
cat(sprintf("se_d                   = %.3f\n", se_d))
cat(sprintf("t_{0.025, %d}           = %.3f\n", df_p, CritVal))
cat(sprintf("95%% CI = (%.3f , %.3f)\n", CI[1], CI[2]))
cat(sprintf("[t.test CI]  (%.3f , %.3f)\n",
            t_paired$conf.int[1], t_paired$conf.int[2]))
# 0 在 CI 裡嗎？順手做一下整體解釋
if (CI[1] <= 0 && 0 <= CI[2]) {
  cat("Interpretation: 0 落在 95% CI 內，沒有證據說兩種語言的平均 coding time 不同。\n")
} else {
  cat("Interpretation: 0 落在 95% CI 之外，有證據說兩種語言平均 coding time 不同。\n")
}

# --- (b) Normality of the differences ---
print_subhead("(b) Normality check on the paired differences")
(sw_p <- shapiro.test(Coding.diff))
save_normality_plots(Coding.diff,
                     "Ex 5-41 Coding.diff",
                     "5-41_diff",
                     x_label = "Coding.diff = Lang1 - Lang2 (minutes)")
# qqPlot 來自 car，老師範例常用
png("figures/5-41_diff_qqplot.png", width = 1100, height = 800, res = 150)
car::qqPlot(Coding.diff,
            main = "Ex 5-41: paired diff Q-Q plot",
            ylab = "Coding.diff (Lang1 - Lang2)")
invisible(dev.off())
cat("Figures saved: figures/5-41_diff_qq.png, figures/5-41_diff_hist.png, ",
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    "figures/5-41_diff_qqplot.png\n", sep = "")


## ---- ex5_59 ----
# ==============================================================================
# EXERCISE 5-59: Equality of Two Variances  (Section 5-5)
# 沿用 5-24：s1 = 22.5, n1 = 10; s2 = 21, n2 = 16; 雙尾 alpha = 0.05
# ==============================================================================
print_header("EXERCISE 5-59: F-TEST FOR EQUAL VARIANCES (GEAR DATA)")

n     <- c(10, 16)
s     <- c(22.5, 21)
alpha <- 0.05

(f0 <- s[1]^2 / s[2]^2)
df  <- c(n[1] - 1, n[2] - 1)

# 雙尾 P-value：兩端取小的乘 2
PVal <- 2 * min(pf(f0, df[1], df[2]), 1 - pf(f0, df[1], df[2]))
# 兩個臨界值，幫 grader 看
f_low  <- qf(    alpha/2, df[1], df[2])
f_high <- qf(1 - alpha/2, df[1], df[2])

print_subhead("Two-sided F-test, alpha = 0.05")
cat("H0: sigma1^2  = sigma2^2\n")
cat("H1: sigma1^2 != sigma2^2\n")
cat(sprintf("f0      = s1^2 / s2^2 = %.3f\n", f0))
cat(sprintf("df      = (%d, %d)\n", df[1], df[2]))
cat(sprintf("F_{0.025, %d, %d} = %.3f , F_{0.975, %d, %d} = %.3f\n",
            df[1], df[2], f_low, df[1], df[2], f_high))
cat(sprintf("P-value = %.3f\n", PVal))
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")

# 也順手丟 95% CI for sigma1^2 / sigma2^2
LB <- (s[1]^2 / s[2]^2) / qf(1 - alpha/2, df[1], df[2])
UB <- (s[1]^2 / s[2]^2) / qf(    alpha/2, df[1], df[2])
cat(sprintf("95%% CI for sigma1^2 / sigma2^2 = (%.3f , %.3f)\n", LB, UB))
cat("1 落在 CI 裡 -> 跟 P-value 結論一致（不拒絕兩變異數相等）。\n")

# F(9, 15) 分布 + 雙尾拒絕區 + f0 標記
f_grid <- seq(0.001, 6, length.out = 700)
f_df_plot <- tibble(f = f_grid, dens = df(f_grid, df[1], df[2]))
rej_lo_f <- f_df_plot %>% filter(f <= f_low)
rej_hi_f <- f_df_plot %>% filter(f >= f_high)
p_f_5_59 <- ggplot(f_df_plot, aes(f, dens)) +
  geom_area(data = rej_lo_f, fill = "firebrick", alpha = 0.45) +
  geom_area(data = rej_hi_f, fill = "firebrick", alpha = 0.45) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_vline(xintercept = c(f_low, f_high),
             linetype = "dashed", color = "gray30") +
  geom_vline(xintercept = f0, linetype = "solid",
             color = "darkgreen", linewidth = 1) +
  # 三個標籤往畫面右上半排，避開曲線、彼此也不重疊
  annotate("text", x = 2.4, y = 0.78,
           label = sprintf("F_{0.025, 9, 15} = %.3f  (left dashed line)", f_low),
           color = "firebrick", size = 4, hjust = 0) +
  annotate("text", x = 2.4, y = 0.68,
           label = sprintf("F_{0.975, 9, 15} = %.3f  (right dashed line)", f_high),
           color = "firebrick", size = 4, hjust = 0) +
  annotate("text", x = 2.4, y = 0.58,
```

```
          label = sprintf("observed f0 = %.3f  (solid green line)", f0),
          color = "darkgreen", size = 4, hjust = 0) +
  # tick 拆成整數，避免 1.00 跟 1.15 擠在一起
  scale_x_continuous(breaks = c(0, 1, 2, 3, 4, 5, 6),
                     limits = c(0, 6)) +
  labs(title = "Ex 5-59 Two-sided F-test: rejection regions and observed f0",
       subtitle = "df = (9, 15), alpha = 0.05; f0 = s1^2 / s2^2 = 1.148",
       x = "F statistic", y = "Density") +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-59_f_rejection.png", p_f_5_59,
       width = 8.5, height = 5.0, dpi = 220)


## ---- ex5_69 ----
# ==============================================================================
# EXERCISE 5-69: Vehicle Rollover Rates  (Section 5-6)
# Two proportions; n = c(100, 100); x = c(35, 25)
# 助教校正：(b)(c) 的 pA = 0.4, pB = 0.25 是 H1 下的 p (power / sample size 用)
# ==============================================================================
print_header("EXERCISE 5-69: ROLLOVER RATES (TWO PROPORTIONS)")

n     <- c(100, 100)
x     <- c(35, 25)
alpha <- 0.05

st <- TwoPropStats(x, n)

# --- (a) One-sided test: pA > pB ---
print_subhead("(a) One-sided two-proportion z-test (P-value approach)")
(z0    <- st$phat.diff / st$se.p)
(PVal  <- 1 - pnorm(z0))
cat("H0: pA  = pB\n")
cat("H1: pA  > pB\n")
cat(sprintf("phat.A   = %.3f , phat.B   = %.3f\n", st$phat[1], st$phat[2]))
cat(sprintf("phat.p   = %.3f , se.p     = %.3f\n", st$phat.p, st$se.p))
cat(sprintf("z0       = %.3f\n", z0))
cat(sprintf("P-value  = %.3f\n", PVal))
cat("Decision: ",
    if (PVal < alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# prop.test 對結果 (correct = FALSE 才能跟 z 公式對到)
pchk <- prop.test(x, n, alternative = "greater", correct = FALSE)
cat(sprintf("[prop.test]  chi^2 = %.3f, p = %.3f\n",
            pchk$statistic, pchk$p.value))

# --- (b) Power at alpha = 0.05, pA = 0.4, pB = 0.25 (H1 下) ---
print_subhead("(b) Power assuming pA = 0.4, pB = 0.25 (H1)")
pA_b <- 0.40
pB_b <- 0.25
pwr_b <- Pwr.prop2.test(n = n, pA = pA_b, pB = pB_b,
                        alpha = alpha, alt = "greater")
cat(sprintf("Under H1: pA = %.2f, pB = %.2f -> pA - pB = %.2f\n",
            pA_b, pB_b, pA_b - pB_b))
cat(sprintf("Power = %.3f , beta = %.3f\n", pwr_b, 1 - pwr_b))

# --- (c) 樣本數夠不夠 ---
print_subhead("(c) Is n = 100 each enough for power >= 0.90?")
n_req <- SplSz.prop2.test(pA = pA_b, pB = pB_b,
                          alpha = alpha, pwr = 0.90, side = 1)
pwr_at100 <- Pwr.prop2.test(n = c(100, 100), pA = pA_b, pB = pB_b,
```

```
                              alpha = alpha, alt = "greater")
cat(sprintf("Required n (each group) = %d\n", n_req))
cat(sprintf("Power at current n = 100 each: %.3f\n", pwr_at100))
cat(" 結論：n=100 還沒到 0.9 power，按公式得補到 ", n_req, " 每組才夠。\n", sep = "")

# Power 對「每組樣本數」的關係圖：把現有 n=100 跟需要的 n=166 都標出來
n_grid <- seq(30, 250, by = 2)
pwr_grid_5_69 <- tibble(
  n_each = n_grid,
  power  = vapply(n_grid,
                  function(nn) Pwr.prop2.test(c(nn, nn),
                                              pA = pA_b, pB = pB_b,
                                              alpha = alpha, alt = "greater"),
                  numeric(1))
)
p_pwr_5_69 <- ggplot(pwr_grid_5_69, aes(n_each, power)) +
  geom_line(color = "steelblue", linewidth = 1.1) +
  geom_hline(yintercept = 0.90, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = 100,   linetype = "dotted",  color = "darkgreen") +
  geom_vline(xintercept = n_req, linetype = "dotted",  color = "purple") +
  geom_point(aes(x = 100,   y = pwr_at100), color = "darkgreen", size = 3) +
  geom_point(aes(x = n_req, y = 0.90),      color = "purple",    size = 3) +
  annotate("text", x = 100, y = pwr_at100 - 0.08,
           label = sprintf("n = 100,\npower = %.3f", pwr_at100),
           color = "darkgreen", size = 3.8, hjust = 1.05) +
  annotate("text", x = n_req, y = 0.82,
           label = sprintf("n = %d for\npower >= 0.90", n_req),
           color = "purple", size = 3.8, hjust = -0.05) +
  scale_y_continuous(breaks = seq(0, 1, 0.1), limits = c(0, 1.02)) +
  labs(title = "Ex 5-69 Two-proportion test: power vs per-group sample size",
       subtitle = "Under H1: pA = 0.40, pB = 0.25; alpha = 0.05; one-sided",
       x = "Per-group sample size  n",
       y = expression(paste("Power  = 1 - ", beta))) +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"),
        axis.title = element_text(face = "bold"))
ggsave("figures/5-69_power_vs_n.png", p_pwr_5_69,
       width = 8.0, height = 5.0, dpi = 220)


## ---- ex5_71 ----
# ==============================================================================
# EXERCISE 5-71: 95% Lower Bound on pA - pB  (Section 5-6)
# 沿用 5-69 資料；Wald 一邊下界
# ==============================================================================
print_header("EXERCISE 5-71: 95% LOWER CONFIDENCE BOUND FOR pA - pB")

n  <- c(100, 100)
x  <- c(35, 25)
alpha <- 0.05
st <- TwoPropStats(x, n)

LB <- st$phat.diff - qnorm(1 - alpha) * st$sehat
cat(sprintf("phat.diff = %.3f , sehat = %.3f\n", st$phat.diff, st$sehat))
cat(sprintf("z_{0.05}  = %.3f\n", qnorm(1 - alpha)))
cat(sprintf("Lower bound LB = phat.diff - z_{0.05} * sehat = %.3f\n", LB))
cat(sprintf("95%% one-sided lower bound : [ %.3f , 1 ]\n", LB))
# 解釋
if (LB > 0) {
  cat("LB > 0 -> 至少 95% 信心保證 pA > pB；和 5-69(a) 結論一致。\n")
} else {
```

```
  cat("LB <= 0 -> 0 還在區間裡，沒有 95% 信心斷言 pA > pB；跟 5-69(a) 邊界 P-value 一致。\n")
}


## ---- ex5_73 ----
# ==============================================================================
# EXERCISE 5-73: Plus-Four (Agresti-Caffo) Lower Bound  (Section 5-6)
# 沿用 5-69 資料；ntilde = n + 2, xtilde = x + 1
# ==============================================================================
print_header("EXERCISE 5-73: NEW CI (PLUS-FOUR / AGRESTI-CAFFO)")

n  <- c(100, 100)
x  <- c(35, 25)
alpha <- 0.05
st <- TwoPropStats(x, n)

LB_new <- st$ptilde.diff - qnorm(1 - alpha) * st$setilde
# 對照 5-71 的傳統下界
LB_trad <- st$phat.diff - qnorm(1 - alpha) * st$sehat

cat(sprintf("ntilde = (%d, %d)\n", st$ntilde[1], st$ntilde[2]))
cat(sprintf("xtilde = (%d, %d)\n", st$xtilde[1], st$xtilde[2]))
cat(sprintf("ptilde     = (%.3f, %.3f)\n", st$ptilde[1], st$ptilde[2]))
cat(sprintf("ptilde.diff = %.3f , setilde = %.3f\n",
            st$ptilde.diff, st$setilde))
cat(sprintf("New lower bound LB_new  = %.3f\n", LB_new))
cat(sprintf("Traditional bound LB    = %.3f (from 5-71)\n", LB_trad))
cat(sprintf("Difference (new - trad) = %.3f\n", LB_new - LB_trad))
cat(" 解讀：plus-four 把 (x, n) 換成 (x+1, n+2)，等於各組多 1 顆成功 1 顆失敗，\n")
cat("       ptilde 更靠近 0.5，setilde 微幅變大，下界比傳統 Wald 略保守。\n")

# 把兩條單邊下界畫成水平線段，0 標一條垂直參考
lb_df <- tibble(
  method = factor(c("Traditional Wald (5-71)",
                    "Plus-four / Agresti-Caffo (5-73)"),
                  levels = c("Traditional Wald (5-71)",
                             "Plus-four / Agresti-Caffo (5-73)")),
  point  = c(st$phat.diff,  st$ptilde.diff),
  lb     = c(LB_trad,       LB_new)
)
p_lb <- ggplot(lb_df, aes(y = method)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "firebrick") +
  geom_segment(aes(x = lb, xend = point, yend = method),
               linewidth = 1.8, color = "steelblue") +
  geom_point(aes(x = point), size = 4, color = "navy") +
  geom_point(aes(x = lb),    size = 3.6, color = "darkorange3") +
  geom_text(aes(x = lb, label = sprintf("LB = %.3f", lb)),
            color = "darkorange3", vjust = -1.4, size = 4) +
  geom_text(aes(x = point, label = sprintf("point = %.3f", point)),
            color = "navy", vjust = -1.4, hjust = 0.5, size = 4) +
  # 左右各保留留白，避免 labels 被切掉
  scale_x_continuous(breaks = seq(-0.05, 0.15, 0.025),
                     limits = c(-0.03, 0.135),
                     expand = expansion(mult = 0.04)) +
  labs(title = "Ex 5-71 vs 5-73: 95% one-sided lower bound on pA - pB",
       subtitle = "Both bounds straddle 0, so neither can conclude pA > pB at 95% confidence",
       x = expression(paste(p[A], " - ", p[B], "  (percentage points)")),
       y = NULL) +
  theme_minimal(base_size = 13) +
  theme(plot.title  = element_text(face = "bold"),
        axis.title  = element_text(face = "bold"),
```

```
        axis.text.y = element_text(face = "bold", size = 11),
        plot.margin = margin(10, 18, 10, 10))
ggsave("figures/5-71-73_lb_compare.png", p_lb,
       width = 9.0, height = 4.2, dpi = 220)


# ==============================================================================
# 結束
# ==============================================================================
cat("\n============================================================\n")
cat(" All Chapter 5 exercises completed.\n")
cat("============================================================\n")
```